

FRM 核心考点速通笔记

FOCO 备考速通系列

说明

FRM 考试的主要参考题目是 3 套 PE 以及机经回忆，基于这些题目，我们整理了相应的考点和保姆级的习题详解，希望这份笔记能够帮助大家快速掌握 FRM 考试的核心考点。这份笔记特点如下：

- (1) 在教材整体框架下梳理考点
- (2) 覆盖 PE1、PE2 和 PE3 的 300 道题的考点
- (3) 同时也覆盖近两年的考经考点
- (4) 每个习题都有详细的解析，尽可能做到保姆级地讲解
- (5) 高度浓缩，全是重点
- (6) 单栏排版，80g 护眼纸
- (7) 附有考点目录和 PE 习题索引目录，查找方便
- (8) 可以补充相关要点在上面，形成自己的备考笔记

FRM 备考建议：

1. 聚焦高质量的习题，并结合考经和机经进行复习
2. 注重知识点的理解，而不是死记硬背

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Operational Risk Measurement and Management

操作风险计量和管理

1 Operational Risk and Resilience 相关概念

考点 1 Operational Risk and Resilience 相关概念

1 操作风险定义

- 运营风险被定义为因内部流程、人员和系统不足或失效或外部事件造成损失的风险。
 - 该定义包括法律风险，但不包括战略和声誉风险。
 - 然而，RPSMOR 于 2021 年发布：在适当情况下，银行的操作风险管理应考虑战略和声誉风险。

2 操作风险分类

- 内部欺诈（Internal fraud, IF）
 - 内部欺诈是指公司员工实施或企图实施的欺诈行为。包括两个分项：
 - * 未经授权的活动
 - * 盗窃和欺诈（包括敲诈勒索、挪用公款、侵占资产、贿赂等）
- 外部欺诈（External fraud, EF）
 - 外部欺诈是指第三方或外部人员对企业实施或企图实施的欺诈行为，常见于零售企业。包括两个子项：
 - * 盗窃和欺诈（如支票拼凑和伪造）
 - * 系统安全（如黑客攻击）
- 就业政策和工作场所安全性（Employment practices workplace safety, EPWS）
 - 与欧洲或亚洲相比，美洲的就业惯例和工作场所安全（EPWS）类型更为突出，因为那里的劳动法比较陈旧，更多的是针对雇主的诉讼文化。包括三个分项目：
 - * 员工关系
 - * 安全环境
 - * 多样性和歧视
- 客户、产品和业务操作
 - 与客户和交易对手发生纠纷、因不当商业行为或错误咨询活动而被监管机构罚款所造成的损失。包括：
 - * 错误信息
 - * 投诉（complaints）
 - * 产品规格等错误招致的折扣贬损
- 实物资产损害（Damage to physical assets, DPA）
 - 涉及所有与自然灾害有关的损失事件以及来自恐怖主义和人为破坏等来源的人为损失（故意破坏公共财物）。
 - 这种风险最常用的评估方法是利用保险进行情景分析。
 - * 很少有公司主动收集这类风险的损失，因为这些损失通常要么太小，要么大得惊人
- 业务中断和系统出错（Business disruption and system failures, BDSF）
 - 业务中断或系统故障造成的损失最难发现。
 - * 例如，IT 故障、停电

- 执行、交割和流程管理
 - 交易处理失败造成的损失，以及与交易对手和供应商之间的问题
 - * 例如处理错误、文件缺失、供应商纠纷等。

3 操作风险频率

表 1: Event Categories with Frequency and Severity

Event category	Frequency	Severity
Internal fraud - IF	2%	2%
External fraud - EF	30%	9%
Employment practices and workplace safety - EPWS	15%	5%
Clients, products, and business practices - CPBP	22%	52%
Damage to physical assets - DPA	1%	1%
Business disruption and system failures - BDSF	2%	5%
Execution, delivery, and process management - EDPM	28%	27%

- CPBP 类别占不到四分之一事件频率造成的运营损失的一半以上。
- EF、EPWS、CPBP 和 EDPM 占运营风险事件频率的 75
- CPBP 和 EDPM 足以解释 79% 的严重程度。
- 值得注意的是，报告中的数字只反映了与每起操作风险事件相关的直接经济损失。
- 大型操作风险事件的间接后果，如声誉受损、市值下降、管理层更替和补救成本，有时远远超过直接财务成本。

4 操作风险的特点

- 异质性 (Heterogenous)
 - 操作风险是一系列不拘一格的风险，其成因和后果各不相同。即使在同一风险类别中，操作风险事件也可能大相径庭。
- 偶发性和扩散性
 - EDPM 等操作风险类型由公司流程和系统的质量驱动或缓解。其他类型如 DPA 通常由外部事件引起。
- 重尾 (heavy tailed)
 - 业务风险以大量小额损失或少量巨额损失的形式出现。
- 相互关联
 - 不同类型的操作风险相互关联，因为它们可能有共同的内部原因，如内部控制薄弱、人为错误或风险文化不佳等。
- 动态
 - 许多业务风险的性质和强度取决于组织或行业的活动，并随着这些活动而变化。操作风险的演变与行业本身的发展相一致。

5 操作风险管理 (ORM) 框架

- 代表所部署的行动、技术或工具。
- 无论采用哪种表述方式，风险管理框架都将范围缩小到四项主要活动：
 - 风险识别 (risk identification)
 - 风险评估 (risk assessment)

- 风险缓解 (risk mitigation)
- 风险报告 (risk monitoring)
- 这四项活动通常以一个连续的循环来体现，以适应风险管理的迭代性质。
- 风险管治涉及执行和报告风险管理的职责分配。
- 风险文化和行为与获得每个人对风险管理和良好行为的认同有关。
- 风险偏好和容忍度是对受监管金融服务公司的一项要求，以确定其在追求目标的过程中愿意面对的风险限度。

6 运营复原力 (Operational resilience)

- 2010 年代后期，当金融部门和监管机构认为不仅需要关注金融稳定性，还需要关注运营稳定性，以及预防和从运营中断中恢复的能力时，对复原力的关注与日俱增。
- 这种关注源于金融业对 IT 系统、基础设施和第三方供应商的严重依赖。
- 英国发布首份运营弹性监管指南。英国金融行为监管局 (FCA)、审慎监管局 (PRA) 和英格兰银行 (BoE) 将运营复原力定义为：
 - 企业和整个金融部门预防、适应、应对、恢复业务中断并从中学习的能力。
- 运营复原力的要素：
 - 业务服务的连续性：降低关键业务供应中断的风险。
 - 重要（关键）业务服务：一旦中断会对消费者或市场完整性造成不可容忍的损害的服务，并确保这些服务的连续性在可容忍的范围内。
 - 影响容忍度：发生事故时可容忍的中断程度，类似于业务连续性规划中的恢复时间目标。
 - * 影响容忍度必须包括基于时间的指标。
 - * 容忍水平旨在帮助高级管理层和董事会制定自己的运营恢复能力标准，确定优先次序，并作出投资决定。
 - * 影响容忍度的设定应假定会发生中断。
 - * 在设定单项重要业务服务的影响容忍时，企业应考虑到其他相关重要业务服务故障的影响。
 - 中断管理：应对中断、维护主要利益相关方的信任以及在危机时刻保持沟通的清晰度，是促进系统稳定的重要因素。
 - 吸取教训：危机过后，可以从内部事故或其他公司的公共事件中吸取教训。
 - 2020 年，美联储发布了《加强业务复原力的稳健做法》，并将业务复原力视为 ORM 不同组成部分的整体结果。
 - ORM 是运营复原力的核心组成部分。
 - ORM 有两个专业领域：第三方风险管理，这对供应链的复原力至关重要；情景分析，这是对尾随事件的必要准备。
 - 业务连续性管理 (BCM) 和 IT 系统复原力是运营复原力的两个关键支持支柱。

考点 2 Risk Governance

1 巴塞尔银行委员会健全 ORM 的原则

- P1: Culture led by the board of directors and implemented by senior management.
- P2: Maintenance of a sound and proportionate ORM framework (ORMF).
- P3: Board review and approval of ORMF.
- P4: Risk Appetite and tolerance statement for operational risk to be approved and periodically reviewed by the board.

- P5: Senior management role in ORM policies and systems development and implementation.
- P6: Comprehensive identification and assessment of operational risk in material activities.
- P7: Change management process adequately resourced and articulated.
- P8: Regular monitoring of operational risk profile and exposures.
- P9: Strong control environment: internal controls, mitigation, training, and risk-transfer strategies.
- PIO: Robust information and communication technology (ICT) management program, in line with ORMF.
- PII: Business continuity plans in place and linked with ORMF.
- P12: Public disclosures on approach to ORM and risk exposures.

2 Supervisory expectation on governance of ORM

- The fundamental expectation from regulators is that ORM should not be a mere compliance exercise of policies on paper, rubber-stamped by the governing body.
- To protect against this risk of non-compliance, it is good practice to maintain a library of regulatory documents that govern how the operational risk framework should operate and ensure that all relevant staff are aware of their contents.
 - Operational risk staff should be asked to confirm annually that they are fully aware of listed documents and contents.

3 Committee structure in risk governance

- The lowest level oversees activity by business or function.
- The group operational risk committee oversees, manages, monitors, and reports the consolidated picture to executive/management committee and board risk committee.
- The board risk committee oversees all operational risk.
- Operational risk is often split into several risk categories, with each category having its own committee in many large organizations, subordinated to:
 - either the board risk committee (or enterprise risk committee),
 - or to the operational risk committee, reporting in turn to the enterprise risk committee.

4 Role of BOD in risk governance

- The board of directors is ultimately responsible for:
 - general administration of the firm.
 - setting the risk appetite of the firm and for making sure that it operates within the limits of its risk appetite.
 - establishing a risk management culture.
 - validating the operational risk management framework and ensuring a periodic revision of the ORMF.

5 Three lines of defense model

- Line 1: This is the risk owners understood as the front line of the business, so-called business unit management.
- Line 2: This is an independent corporate operational risk management function (CORF) which provides guidance, oversight and challenge of the first line.
- Line 3: This is the internal audit function serving as an independent assurance challenging line 1 and 2. Internal audit should report in solid line to non-executive directors.

- Difference between line 1 and line 2: The first line owns the final sign-off on risks assessment and controls, this is not the role of the second line. The first line owns the risks and the second line owns the methodology.
- Line 1.5: Risk specialists (risk champions or risk correspondents or stewards) in each business department to interact with the risk function which are particularly common in larger organizations.

6 Risk Appetite

- Definition: The board is responsible for determining and defining the nature and extent of the significant risks it is willing to take in achieving its strategic objectives.
- It is the role of the board of directors to ultimately own the risk limits and to validate them periodically.
- In practice, the board often delegates this responsibility to the board risk committee, which in turn requests the support of the risk function to ensure the implementation of these limits in controls and exposure in the business, and to regularly report on them.
- Risk appetite governance
 - Allocate a risk owner to each risk type labeled in the risk appetite structure.
 - Designate control owners and metrics owners for the various operational risks when the risk appetite subcategories and controls cascade down into limits, controls, and monitoring metrics.
- Risk owners are responsible for managing, maintaining, and monitoring the risks with the stated limits of appetite and tolerance. Risk owners are the first line of defense.
- Controls owners are responsible for the design, implementation, and effectiveness of the controls.
- Metrics owners are responsible for the collection, reporting and monitoring of the metrics capturing the performance of the organization with regards to risk appetite.
- Regulatory expectations for risk appetite
 - Easy to communicate and understand.
 - In alignment with the bank's strategy, business plans and operations of the firms.
 - Risk limits should be monitored through indicators related to risk exposure, control environment, and loss events.
 - Risk appetite should be forward-looking and subject to stress testing and have a view of what events might push the bank outside the limits of risk appetite and tolerance.

7 Definition of culture

- Culture is the summation of values and ethics, desired conduct standards, and implied behaviors. While cultural norms and beliefs can't easily be measured, the conduct and behaviors that the culture norms can be observed, monitored, managed and incentivized. Conduct and behaviors are the only visible elements of culture.

8 Risk culture

- In the eyes of regulators, risk culture is closely associated with good conduct and ethics, and falls under the responsibilities of the board.
 - Principle 1 by BCBS: The board of directors (BOD) should take the lead in establishing a strong risk management culture.
- The board of directors must lead the risk culture to be implemented by senior management. This is referred to as "tone at the top".

- Risk culture is supported by management reinforcement of:
 - Codes of conduct and ethics.
 - Compensation structure.
 - Training to raise awareness of operational risks.

9 Risk culture-Codes of conduct and ethics

- Reinforcement of adherence to the code of conduct and discipline are effective ways to influence behaviors.
 - But unnecessary blame should be avoided. A "no-blame culture" along with "tone from the top" is the most common recommendation for the establishment of a good risk culture.

10 Risk culture-Compensation structure

- Toxic risk cultures are often triggered by overambitious shortterm growth and profit targets, undue pressure on staff and management, and resistance to bad news.
 - Aggressive compensation structure like that the bank increases the proportion of incentive compensation for its traders and investment bankers should not be recommended.

11 Risk culture-Training

- Common risk training programs include:
 - induction training for new hires
 - online modules available to all staff
 - specialized training for the risk function

12 Signs of good risk culture

- By experience, there are a handful of reliable signs of a healthy risk culture in organizations:
 - applicability of the rules to everyone
 - swift escalation of issues
 - sharing of the lessons learned

考点 3 Risk Identification

1 Scope of risk identification

- Top-down: Start from the senior leadership, then down to business units, and finally to individual business processes.
- Bottom-up: It complements top-down approach. Businesses and divisions evaluate their own operational risk.
- 通过自上而下和自下而上的协调，可以全面了解操作风险状况。

2 Top-down

- Target: Aim at identifying key organizational risks, the major business threats that could jeopardize strategic objectives.
- Outcome: Risk prioritization and risk ranking is an important outcome of top-down risk identification.
- Tools:
 - Business-specific risk identification
 - The risk wheel

- Emerging risk identification: horizon scanning
- Tools: Business-specific risk identification: exposures and vulnerabilities
 - Exposure drives impacts in risk, should a failure materialize in one of these activities.
 - Vulnerabilities are the weakest links in business activities. These can be inadequate or outdated products, systems overdue for maintenance and testing, non-core businesses left unmonitored.
 - If vulnerabilities are aligned with large exposures, this can result in very significant losses or a failure of the business.
 - The benefits of listing exposures and vulnerabilities as a risk identification brainstorming method has the main advantage of being business specific.
- Tools: The risk wheel
 - The risk wheel is a classic brainstorming tool to encourage creativity and ideas during risk identification workshops.
 - Domino effect: Salary cut down □ personal effectiveness □ delays or descoping of project IT upgrade □ IT risk materializes □ systems disruption □ impact business continuity □ reputational damage.
- Tools: Emerging risk identification: horizon scanning
 - Known emerging risks: such as cyber, compliance, and climate change. In this, "emerging" means "increasing."
 - Nascent emerging risks: just starting to appear on the risk radar of a community or of an organization and truly deserve the term of emerging risk.
 - Benchmarking pieces of information can be used for inspiration or as a checklist.
 - A way of scanning horizon risks is PESTLE analysis:
 - * Political
 - * Economic
 - * Social
 - * Technological
 - * Legal
 - * Environmental

3 Bottom-up

- Bottom-up risk identification tools
 - Event and loss data analysis
 - Risk and Control Self-Assessment (RCSA)
 - Process mapping
- Event and loss data analysis
 - Internal losses: Repeated internal losses might be a sign of failure in the internal controls system.
 - External losses: Such as consortium data. A best practice is to monitor all large incidents communicated by peers and ask objectively: "Could this incident happen to us?"
 - Near misses: Incidents that could have resulted in an operational loss but did not because of good luck or outside intervention. Near misses are "lessons for free".
- Risk and control self-assessment (RCSA):
 - An organization, a business line, or a department evaluates the likelihood and the impact of its operational risks. This gives information on the level of residual risks.
 - RCSA is often workshop-style discussion facilitated by the second line to collect information from the first line.
 - In the largest institutions, the process can take the form of questionnaires.

- Selecting the relevant participants: Two types of employees are suitable to interview. One group is the most experienced employees. The other group comprises recent hires.
- Frequency of RCSA: typically performed annually, updated after each trigger event in mature firms. If performed too frequently, the process can turn into a box-ticking exercise.
- The choice of the granularity of scope of RCSA workshops
 - * If too granular, the output will be a collection of small daily risks and inefficiencies that are not always of much value. People may miss the big picture when it comes to risks.
- Process mapping: Lay out the tasks of a process, step by step, and asking what can go wrong in each step. This exercise highlights possible under-control or over-control of risks.
- For process mapping, appropriate analysis is important:
 - If too granular, the process mapping will be excessively time-consuming and likely to raise only minor issues.
 - If too high-level, it won't be informative enough to be used.

4 Scenario analysis- Brainstorming techniques

- The regulatory guidance document describes scenario analysis as an exercise of extreme events identification, using workshops and brainstorming techniques.
 - Brainstorming techniques are a method to identify, analyze and measure a range of scenarios, including low probability and high severity events.
- The preparation phase: Firms are required to minimize subjective biases when conducting scenario analysis.
 - The participants in scenario analysis workshops should be senior managers within the different business functions with experience and understanding of the risks.
 - The involvement of additional external experts is advisable to help mitigate some biases such as myopia, excessive focus on scenarios driven by external causes.
- The generation phase aimed at producing a long list of scenarios to be considered.
- The intermediary phase: Scenario selection in which some scenarios are consolidated, and others eliminated or added.
 - An example of consolidated scenarios are those relating to the same internal impact but different external causes.
- It is useful to compare the bank's scenarios with an industry scenarios, to determine whether omitting relevant scenarios.
- Some scenarios can be eliminated if the risk owner can convincingly demonstrate that the maximum loss generated by a given scenario is moderate enough to be absorbed by operating margin and without significant disruption.

5 Examples of risk taxonomy

- Basel taxonomy
 - Level 1 is the highest-level category; level 2 is a detailed version of level 1; level 3 provides examples. The BCBS only recognizes the first two levels as regulatory categories.
 - The number of risks to assess, manage, and report needs to be kept at a reasonable level. Too much detail is detrimental to the quality of information, difficult to review, and can drain resources.
- Operational risk taxonomy by ORX
 - The taxonomy revises the Basel operational risk taxonomy.
 - The taxonomy does not radically depart from the Basel categories but offers some noticeable modifications:
 - * It presents 14 level 1 risk types compared to 7 from Basel, and some level 2 risks have been elevated to level 1 given their prominence in today's world.

6 Structure of taxonomies

- Best practice is to distinguish and separately categorize the different components of uncertainties: causes, risks, impacts, and controls which are organized in taxonomies.
- A structure of taxonomies of causes, controls, risks, and impacts, circling the actionable part of risk management: acting on the causes and the controls.
 - Causes and controls are the prime objects of risk management actionability.
- It is neither reasonable nor economically efficient to attempt to eliminate every risk. However, trying to eliminate the downside of a risk while preserving its upside makes sense.
- To limit the downside of operational risk, organizations and individuals can, first, act on their causes and, next, apply controls on the remaining causes, to limit the two dimensions of a risk-the likelihood and the impact.
- Cause taxonomy
 - The natural level 1 causes for operational risk events by BCBS are PPSE, which is most common method adopted by the market.
 - But not every bank refines the categories to finer level 2 which provides useful insights into the causes of incidents.
- Impact Taxonomy

表 2: ORX Reference Impact Taxonomy

Level 1	Level 2
Direct financial impact	Such as timing loss, legal cost, damaged assets
Indirect financial impact	Such as opportunity costs, disruption
Non-financial impact	Stakeholder groups: customers, employees, third parties (vendors, suppliers, and service providers), shareholders, market/competition

- Control taxonomy
 - Preventative control
 - Detective control
 - Corrective control
 - Directive control

考点 4 Risk Measurement and Assessment

1 Loss data collection

- In 2000, ING created its first representation of the ORM framework. It comprised four concentric circles:

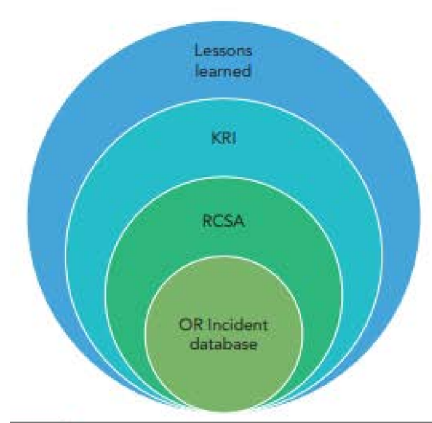


图 1: ORM framework

2 Incident database

- Incident data collection (internal and external)
- Comprehensive data
- Dates of incidents and settlement lags
- Data quality requirements

3 Comprehensiveness of data

- Basel committee does not define "material activities and exposures," but sets a minimum threshold for loss reporting at €20,000 (about \$22,000) .
- From a regulatory perspective, the choice of the reporting threshold should not affect the credibility of the process nor impair management information.
 - It must be justified and should not suggest "cherry-picking " or manipulation of capital requirements.
- From a regulatory perspective, firms must only report incidents leading to financial losses.
- From a management perspective, it is good practice to record also the so-called "non-financial impacts," such as reputational damage, customer detriment, disruption of service, or management time and attention.
- "Non-financial" impact is particularly misleading because the indirect consequences of many material operational risk events have financial implications: Regulatory scrutiny, customers' dissatisfaction, remediation plans, and management attention all have costly consequences.
 - When these repercussions are not properly quantified, the cost of operational risk can be significantly underestimated.
- Grouped losses are distinct operational risk events connected through a common cause.
 - Examples of grouped losses: the same wrong advice given to a group of clients, leading to a series of compensation claims.
- Different events that materialize from a single failure must be grouped as a single loss to reflect the same underlying cause.

4 Dates of incidents

- Date of occurrence: when the event first happened
- Date of discovery: when it is first identified
- Date of reporting: when it enters the reporting database

- Date of accounting: when the financial impact enters the GL
- The difference between occurrence and discovery reflects the visibility of issues, whereas the difference between discovery and reporting shows how diligently operational incidents are reported to the risk function.
- Include a maximum allowed time for reporting incident.
- Best practice dictates a risk-based approach that will not waste resources by rushing to report immaterial events.
 - Material incidents must be reported within a few working days, while minor incidents can be included in periodic summary reporting.
- Reconciling the loss data recorded as operational losses with information from other sources. These sources often include the general ledger (GL) that captures all cash inflows and outflows of a firm, including unintentional losses and gains.
- GL is no substitute for a dedicated operational event database because it only captures the direct financial losses or accidental gains, but it does not link the indirect effects.
- Underreporting is a challenge. Corporate incentives to report operational incidents range from soft encouragement to stringent oversight.
- Self-reporting of incidents is required, apply disciplinary action if unreported events are subsequently discovered. Most organizations include risk metrics in the balanced scorecard of their managers, as part of a risk-based performance system, a practice encouraged by regulators.

5 Qualitative risk assessment

- Risk and Control Self-Assessment (RCSA)
- Key performance indicators(KPIs), Key risk indicators(KRIs), Key control indicators(KCIs)

6 Risk and Control Self-Assessment (RCSA)

- The RCSA refers to evaluating the likelihood and impact of operational risk faced by a business including both the firm's risk exposures before the impact of existing controls has been assessed (inherent risks), and those that remain after the evaluation of the controls in place (residual risks).
- Variations on the RCSA include the risk and control assessment (RCA) and the residual risk self-assessment (RRSA).
- RCSAs are qualitative and largely judgement-based.
- RCSAs are often conducted as workshop-style discussions.
- Some firms may form a dedicated risk assessment unit (RAU), responsible for the conduct of RCSAs. An RAU can be a part of a business line, division, or an organizational process.
- Mature organizations back-test the results of risk assessment against past incident experience.

7 Severity assessment: impact scales

- Four types of impacts: financial, regulatory, customer, and reputation.

8 Likelihood assessment scales

- Likelihood scales are either expressed in percentages of likelihood or frequency of occurrence ("once in x years") .

表 3: Frequency of Occurrence and Probability Assessment

Qualitative Rating	Frequency of Occurrence	Probability of Occurrence % (one year horizon)
Likely	Once per year or more frequently	> 50%
Possible	> 1-5 years	20-50%
Unlikely	1 in 5-20 years	5-20%
Remote	< 1 in 20 years	< 5%

9 Combining likelihood and impact: heatmap

- The various combinations of impact and likelihood are attributed colors that reflect the intensity of the risk. The colors used are red-amber-green or red-amber-yellow-green.

Likelihood				
Likely				
Possible				
Unlikely				
Remote				
	Low	Moderate	Major	Extreme
Impact				

图 2: heatmap

10 Outcome of RCSA

- The results of assessments must be documented and submitted to business line managers for approval and sign off.
- Action plans are owned by the first line (business-line managers) rather than the second line. Line managers are responsible for implementing mitigating action plans.
- Line managers and risk managers could expect to get from an RCSA exercise a view of residual exposure to key risks, when risks assessed as such are deemed above risk appetite, they will be the object of an action plan.

11 Challenge of RCSA

- The first challenge: Subjectivity of judgments, behavioral biases, and limited data input can quickly impede the quality and reliability of output.
- The second challenge: Difficulty in achieving comparable results across different assessment units, different departments, and businesses, often assessed by different people.

12 Key Performance Indicators (KPI)

- KPIs measure performance or target achievement.
- Examples of KPIs include:
 - Maximum downtime/uptime of IT systems
 - Error rates on retail transactions
 - Client satisfaction score/Level of client complaints

13 Key Risk Indicators (KRI)

- KRIs are monitoring metrics that signal the increase/decrease in the level of risk, either in potential impact or in likelihood.
- Examples of KRIs used in measuring likelihood include:
 - Increase in number of transactions per staff (risk of errors)
 - Drop in customer satisfaction scores (risk of client attrition)
 - Increase in the level of sales required for sales staff to achieve a performance objective (risk of fraud)
- Examples of KRIs measuring impact include:
 - Increase in the level of proprietary company knowledge held by a key employee (higher impact of discontinuity/loss of knowledge in case of departure/ sickness)
 - Increase in sensitivity of data held on given server (higher impact in case of data leakage/loss)
 - Increase in value generated by top-10 clients (higher impact in case of client attrition)

14 Key Control Indicators (KCI)

- Key control indicators (KCIs) measure the effectiveness of controls, either in design or performance. Examples of KCIs:
 - Missed due diligence items
 - Lack of segregation of duties
 - Overlooked errors in four-eye checks (reviews by two competent individuals)

15 Interaction among KPIs, KRIs, and KCIs

- KPIs, KRIs, and KCIs overlap in many instances, especially when they signal breaches of minimum standards.
- The failure of a control function is jointly a KPI, a KRI, and a KCI.
 - A poor performance will often become a source of risk.
 - A key control failure always constitutes a source of risk.
- KRI thresholds and governance reflect the risk appetite and tolerance levels of an organization.

16 Quantitative risk assessment

- Causal analysis of scenarios
- Fault Tree Analysis (FTA)
- FAIR methodology
- Root-cause analysis and Bowtie tool
- Swiss cheese: control layering

17 Causal analysis of scenarios

- Causal analysis is used to quantify operational risk scenarios, which focuses on tails (rare events) that affects the firm significantly.
- Causal analysis is based on forward-looking (not past) determinants of impacts and likelihood of operational risks.
- Quantitative risk assessment is important but challenging task in risk management when it comes to scenarios.

18 Fault Tree Analysis (FTA)

- Fault trees decompose failure scenarios into external conditions and internal conditions.
- FTA is built upon a series of conditions that can either happen simultaneously (known as "AND conditions") or alternatively (known as "OR conditions").

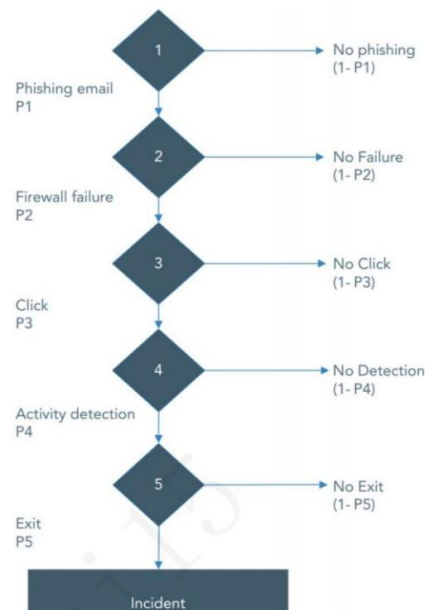


图 3: FTA

- If conditions are independent, likelihood of this risk materializing is: $P1 * P2 * P3 * P4 * P5$.
- If control failures and other impact factors are dependent: Bayesian (conditional) probabilities.
 - Bayesian models refer to measurement methods where likelihood assessments are updated by new information.

19 FAIR methodology

- FAIR (factor analysis of information risk) is a factor model for quantifying operational risk, which is based on decomposition of the risk into their factors.
- The steps are as follows:
 - Identification of the risk factors and interrelationships
 - Measurement/metric of each factor
 - Computational combination of the factors
- In this method, a scenario should have:
 - an asset at risk (asset of the scenario)
 - a threat community (who or what the threat is)
 - a threat type (nature of the threat)
 - an effect (losses resulting from materialization of the risk)

20 Root-cause analysis and Bowtie tool

- It takes the form of the "5-why" analysis. Recommendations require the first line to perform root-cause analysis. The investigations should be challenged by the second line.

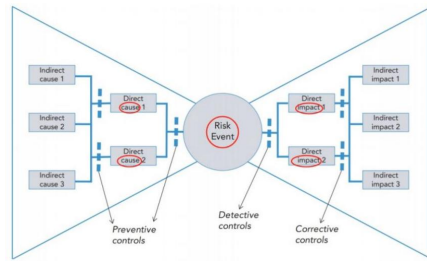


图 4: RISKEVENT

- Bowtie analysis is useful for:
 - assessment of the likelihood and impact of risks.
 - better understanding of the causes of risks and the possible failure of controls that may lead to their materialization.
 - investigating how quickly an incident is detected and how well the organization reacted in limiting the impacts.

21 Swiss cheese: control layering

- It is also called the cumulative act model. Each defensive layer are like slices of Swiss cheese, having many holes, which are continually opening, shutting, and shifting their location. The presence of one hole does not cause a bad outcome but does when holes in many layers momentarily line up.

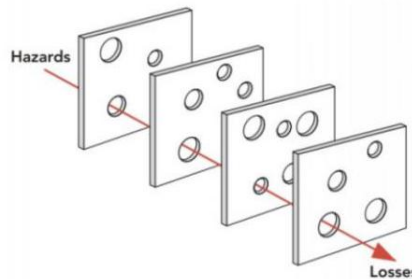


图 5: Swisscheese

- In an effective control layering, where the holes are not aligned (failure rates of controls are independent), and controls compensate each other's weaknesses to produce a control system that works as a tight safety net, rather than like a row of dominos.
- Assessing independence of controls is at least as important as assessing reliability of each individual control.

22 Operational risks modeling and capital

- Loss Distribution Approach (LDA): Advanced Measurement Approach (AMA)
 - Frequency (discrete) is modeled with Poisson distribution and negative binomial distributions.
 - Severity (continuous) is modeled with log-normal distribution, Weibull and Generalized Pareto Distributions (GPD) (More heavy-tailed).
 - Regulators require clear rules to be set for the selection of data and avoid "cherry picking" and manipulation of results.
 - Internal loss data is the backbone of LDA because of more relevant to the organization.
 - External loss data from peers can be supplemented.
 - Mix internal and external data: Scale, Cut-off mix, Filter.
- Limitations and constraints of LDA

- The distributions of severity and frequency of operational losses are typically assumed to be independent.
- Data are split into risk classes (or units of measure), where losses within each risk class should be iid (independent and identically distributed).
- LDA modeling Pillar 1 of regulatory capital are now being transferred to modeling Pillar 2 under ICAAP (Europe), as well as stress-testing calculations (USA) and operational resilience.
 - The ICAAP is part of Pillar 2 of regulatory requirements which covers all financial and non-financial. So, ICAAP applies to all risks, not only operational risk.
 - The ICAAP can inform the economic capital and both have the same purpose.

23 Quantification of operational resilience

- Differences between resilience frameworks and traditional business continuity management (BCM)
 - BCM focuses on the continuity of each business process and recovery, in case of disruption within a given time frame (Recovery Time Objective, RTO).
 - Resilience focuses on the continuity and recovery of important business services.
- Key steps for operational resilience

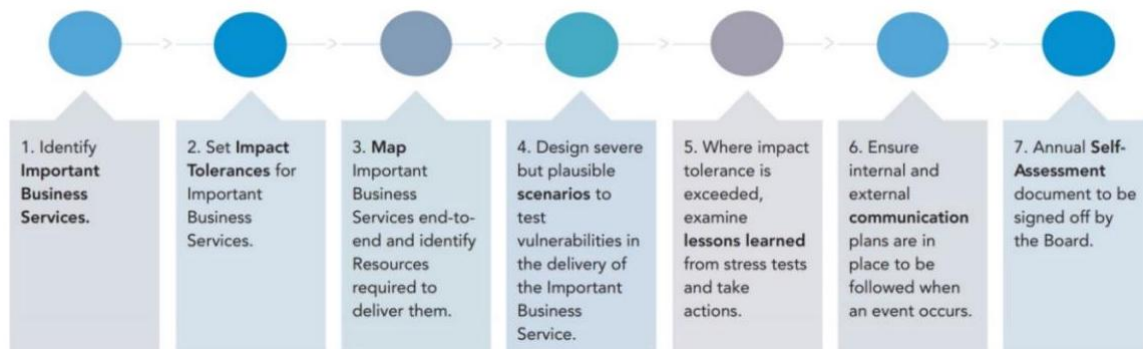


图 6: keysteps

- Impact tolerances can be built up through business impact analysis (BIA). BIA is the assessment of consequences of disruption of a process or a service in business terms, but also on the resources needed to develop recovery strategies.
- The BIA report should prioritize the order of events for restoration. Business processes with the greatest operational and financial impacts should be restored first.
- Single points of failure (SPOF) are relevant KRIs for resilience.
 - One example of KRIs is the number of key employees without back-ups, without documented processes, with unique knowledge and skills.
 - SPOF need to be mitigated and removed through back-up and redundancies, and if not, they need to be considered when setting tolerance thresholds and stress tests.

考点 5 Risk Mitigation

1 Two kinds of operational risk

- Operational risks can be categorized in two types:
 - External operational risks are those that arise due to the business environment of the organization or due to external events. External operational risks cannot be prevented, but they can be reduced.

- Internal operational risks are linked to people, processes, and systems. Internal operational risks tend to be far more preventable, largely through the use of internal controls.

2 Risk response

- Tolerate: accepting the risk as it is.
- Treat: encapsulating all the types of risk mitigation, mostly internal controls.
- Transfer: moving the risk to another party (external insurance).
- Terminate: removal of all risk exposure by discontinuing a product or ceasing operations in certain countries

3 Internal control types

- Preventive controls: intended to reduce the likelihood of an incident.
 - Segregation of duties
 - * The front-office (trading desk) initiates trades, validated by the middle-office, confirmed and settled in the back-office and treasury departments.
 - Access controls (digital or physical)
 - Levels of authorization
 - Process automation
- Detective controls: designed to alert if an incident occurs, accelerate its resolution, and limit the impact of the incident to the firm or its stakeholders.
 - Exception reports
 - File reconciliations
 - Smoke detectors
 - Intrusion detection systems
- Corrective controls: mitigating technique designed to lessen the impact to the institution when adverse events occur. They do not influence the likelihood of a risk happening, but lessen the pain if it does.
 - Data back-ups
 - IT systems patch management
 - Continuity planning
 - Systems redundancies
- Directive controls: include all the prescriptions and rules to execute a process.
 - Procedures and guidance manuals
 - Internal training
 - Team supervision
 - Handover procedures

4 Control design

- Types of weakly designed controls include the following:
 - Optimistic controls
 - Such as sign-offs for large volumes of documents before a deadline.

- Collective controls
Such as "four-eyes check" diluting accountability.
- More of the same
Such as responding to a failure of collective controls by adding more controllers.

5 Control testing

- The main types of control testing are the following:
 - Self-certification or inquiry: limited to secondary controls or controls related to low risk environments.
 - Examination: This requires supporting evidence and documentation. The effectiveness depends on the relevance and adequacy of the documentation. It provides moderate assurance and is better suited for automated controls and sampling of manual controls.
 - Observation: involves real-time oversight of the execution of the control process to judge its design and effectiveness.
 - Reperformance: also called reproduction or parallel testing. The tester reproduces the control process on a sample of transactions and compares the results with those previously obtained by the process. It is the strongest form of testing.

6 Risk mitigation by process design

- Prevention through design
 - Examples include checklists, communication protocols, standardization, and optimized work environments or systems design.
- Typology and mitigation of human error
- Lean Six Sigma
- Quality improvement
- Lean Six Sigma: DMAIC cycle

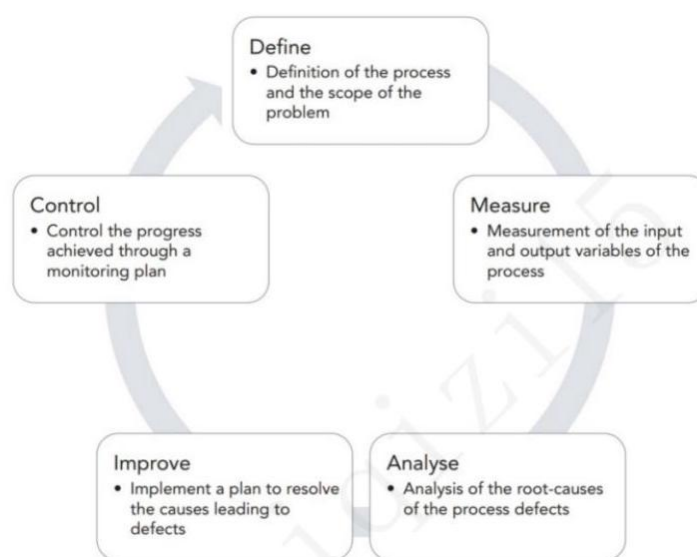


图 7: DMAIC

- Quality improvement: represented by PDSA cycle.
 - Plan: objective; questions and predictions; develop a plan to improve the model (who, what, where, when)
 - Do: carry out the plan; document problems and unexpected observations; begin data analysis

- Study: complete the data analysis; compare data to predictions; summarize what was learned
- Act: what changes are to be made? next cycle?

7 Typology and mitigation of human error

- Four kinds of human error
 - Skilled-based: slip
 - * Involuntary
 - * Caused by inattention, distraction, environment, tiredness.
 - * Controls: Improved, supportive environment, re-engineered processes, adjusted pace, etc.
 - Rule-based: mistakes
 - * Wrong action based on flawed rules
 - * Caused by wrong incentives, flawed products, misleading instructions, etc. One example is mis-selling to customers due to aggressive commercial incentives.
 - * Controls: change the rule
 - Knowledge-based: mistakes
 - * Wrong choice of action facing a new situation
 - * Caused by lack of training, knowledge of the environment, causes and/or consequences of actions
 - * Controls: training, documented procedures, help file, escalation rules
 - A fourth type is a violation
 - * Voluntary misdeed
 - * Controls for violations: Supervisory controls, either human or automated, through hierarchy, cameras, or automated recordings, improvements in risk and compliance culture that reward adherence to rules and processes

8 New Product approval process

- Every project, product, and initiative in a business will bring novelty, lack of familiarity, and uncertainties that may generate significant operational risks.
- A common risk-mitigation approach is the New Product Approval Process (NPAP), and its generalization to new initiatives, New Initiative Risk Assessment Process (NIRAP).
- Best practice requires to present a business case to justify the allocation of resources. A proper business case would cover at a minimum five topics:
 - Objective: what we want the firm to achieve and the rationale for the project.
 - Alternative: Other options considered and reasons for selecting the current project.
- A proper business case would cover at a minimum five topics:
 - Expected benefits: Expected benefits and synergies from the project, and possible drawbacks.
 - Commercial aspects: The costs, investment needs, and funding arrangements.
 - Risks: Main risks and their possible impact on the business case and on the rest of the business, with their mitigation.
- Stages of involvement of risk function during a project's life.
 - Initial stage (before kick-off):
 - * Risk identification and assessment
 - * Mitigation and monitoring plans
 - Project life:

- * monitoring and risk update, regular meetings with the risk team and the project team to update risk identification and assessment findings for important and critical projects.
- Project closure:
 - * Debriefing, evaluations of project deliverables, analysis and documentation of the risks, lessons learned.

9 Mitigation measures to reduce impact

- Key measures of impact reduction:
 - Contingency planning (Plan B)
 - Resilience measurement
 - Crisis management and communication

10 Contingency planning

- Contingency planning is a component of business continuity, disaster recovery, and of corrective risk management, the planning phase of corrective and directive controls.
- Specific forms of contingency planning are business continuity management (BCM) and disaster recovery plans (DRP). These are most relevant when we consider operational resilience
- To be effective, the plan needs to be regularly tested, to ensure the practicality and speed of its implementation in case of an emergency.
- There must be a key owner who has clear responsibility for the design of actions and their execution, including communication with outside parties
- BCM as an ongoing process
 - The first step is to ensure senior level commitment. Topdown direction, support, and ownership are essential in both the BCM process and the quick activation of the BCP when a crisis hits.
 - The next step is to initiate the management process and identify a team to take ownership as a continuing project, rather than a one-off event.

11 Event and crisis management

- To manage a crisis or a large operational event effectively
 - Speed
 - Competence
 - Transparency
- In crisis model, firms should have at a minimum two types of incident response teams:
 - A technical team (or recovery team), composed of specialists who assess the incident and focus on restoring normal processes as quickly as possible.
 - A communication team (external or internal), which deals with the media and the different stakeholder groups, including employees.
- Four typical phases of a major operational risk event:
 - Crisis
 - Emergency response
 - Recovery (RPO and RTO)
 - Restoration (back-to-normal)

- Recovery Point Objective (RPO) indicates the amount of data updated or created that will be lost or need to be re-entered after an outage.
- Recovery Time Objective (RTO) is the amount of downtime a business can tolerate. The RTO is the maximum tolerable duration of disruption for a process, a system, or a service.

12 Risk transfer

- Two methods to transfer risks:
 - External Insurance: An organization agrees to pay a regular premium in exchange for being compensated should a certain risk materialize.
 - Outsourcing: The process of delegating some of the company's tasks to a third party under a contractual agreement, such as IT server management, cloud computing, data centers, or call centers.

13 Risk transfer-external insurance

- For small losses: Many large institutions either self-insure small losses via a captive subsidiary or just absorb the volatility.
- For large losses: Financial institutions mainly use external insurance to cover extreme operational risk "tail" events, such as cyber risks, business discontinuity, or major class action lawsuits.

14 Risk transfer-outsourcing

- For traditional banks: internalize credit decisions and allocations but outsource some of their ICT activities.
- For FinTech banks: more likely to manage their own technology platforms, but outsource credit risk decisions to other specialists.
- Outsourcing can increase or decrease the risk for an organization. depending on what process or activity is outsourced, to whom, and for which reasons.
- Outsourcing can result in increased operational risk labeled third-party risk. Increasingly, outsourcing is seen as a "risk sharing" activity.

15 Management of reputational risk

- Important elements such as reputational damage cannot really be outsourced (nor repaired) through insurance.
- Therefore, prevention and mitigation strategies are essential to build a solid reputation and to react quickly and appropriately in times of crisis to limit reputational damage following operational risk incidents.
- Some preventive controls and strategies are designed to build and maintain customer confidence.
- Detective controls, such as monitoring customer complaints on social media and tracking trends in refund requests or system downtimes, are designed to identify operational failures but also reduce their reputational impact.
- Many corrective risk controls and mitigation strategies are in place to protect the company's reputation and form actions taken when operational events hurt customers.

考点 6 Risk Reporting

1 Roles and responsibilities of committees

- Risk Committee
 - Risk committee is authorized by the board to monitor the effectiveness of risk management framework.

- The risk committee will receive information on key risk indicators associated with the firm’s risk appetite, the frequency and severity of risk events, and the investigation of the causes of large events etc.
- Any significant issues in control system or in operational risk status will be escalated to the risk committee.
- Reporting requirements related to the firm’s operations must be sufficient for the board risk committee to determine that the organization’s control and monitoring systems can effectively comply with, and operate within limits of, the firm’s risk appetite, which is set by the board.
- Executive committee (ExCo)
 - Composed of elected board members and senior executives, acting as a steering committee for board.
 - Facilitate decision-making during times of crisis and prioritize issues for the board to address during normal time.
 - Oversee board policies, and ensure good governance practices.
 - Oversee effective execution of the ORM framework.
- Audit committee
 - Responsible for a third level of operational risk oversight managed by the firm’s internal audit activities (the third line of defense).
 - Responsible for the assurance on the internal control system of the organization.
 - Internal audit and audit committee have several crossovers with respect to ORM activities.
 - * Internal audit function regularly audits operational risk function and report findings to ExCo and audit committee.
 - * Vulnerabilities in the control systems uncovered through internal audits are reported to audit committee.
- Operational risk committee
 - Central ORM function as a second line of defense centralizes most information regarding operational risk and its management.
 - Central ORM function provide an aggregated view of various risks which are often managed in a siloed manner at the business-line level and their interaction.
 - Central ORM function collects all relevant operational risk information from business lines to produce aggregated, synthesized reporting for operational risk committee and provide feedback to business lines, enabling them to:
 - * monitor their own operational risks.
 - * benchmark their performance relative to other business lines or departments, or to a firm-wide average.

2 Considerations of operational reporting

- Reports should be neither too large nor too small.
 - If too large: significant risk of overlooking key pieces of information. Too much information can bury key insights.
 - If too small: it can become meaningless.
- There is little uniformity in operational reporting across firms:
 - Some focus on narrative of past risk events, others focus on forward-looking indicators. The more mature the ORM approach, the more forward-looking its reporting.

3 Seven elements of internal ORM report

- Seven elements included in internal ORM report:
 1. Top-10 risks and risk outlook in risk inventory
 2. Heatmap and risk register
 3. Risk appetite metrics

4. KRIs and issue monitoring
 5. Incidents and near misses
 6. Action plans and follow-up
 7. Emerging risks and horizon scan finding
- Risk appetite metrics
 - Risk appetite metrics (or risk appetite KRIs) are indicators reflecting the firm’s compliance with risk limits for its operational risk appetite.
 - At the business level, these metrics are collected and analyzed individually.
 - When reported to upper levels of management and board, risk appetite metrics are presented as a single list.
 - KRIs and issue monitoring
 - KRIs can provide a granular, forward-looking analysis of risk exposure in different activities.
 - Issues are grouped by business line or a major initiative (new business, merger etc.) to make reporting actionable.
 - Incidents and near misses
 - Increasingly, loss classifications are standardized and shared across financial institutions. One common standard is the ORX global banking loss-data service classification.
 - * Although increasingly standardized, reporting thresholds may vary widely from firm to firm.
 - The materiality of an operational risk event must be judged not only by actual impact but also by potential impact.
 - * Reporting thresholds for risk events and near misses should theoretically be based on potential impact of the event rather than its materialized impact.
 - * Given the difficulty to estimate potential impact, firms and regulators use observed impact to set reporting thresholds.
 - Action plans and follow-up
 - Action plans (including preventive, detective and corrective action plans) are risk-mitigation programs designed to improve the control environment.
 - Action plans are tracked, reported by business-line owner.
 - There is a zero objective for firms where risk management is strong: zero overdue actions and zero overdue audit recommendations.
 - Emerging risks and horizon scan finding
 - In a volatile environment, practice of horizon scanning to identify new trends and emerging risks has been instituted.
 - * Emerging risks are reported regularly to risk committee.
 - * Much of horizon scanning focuses on regulatory risks and changes in the compliance and regulatory environment.
 - * Best practices suggest horizon scanning should reflect elements that could cause change in emerging risks.

4 Characteristics and challenges of operational risk data

- Operational loss data are:
 - heavy tailed or skewed statistically away from the mean.
 - asymmetry with small number of large loss but large number of small losses
- A proper allocation of risk management resources is to focus on the prevention and remediation of large incidents and not to be dispersed and distracted by daily volatility

5 Ways to overcome these challenges

- Escalation of large risk events to upper management.
- Identify and analyze regularly large number of small losses.
- Benchmarking operational losses
 - Reporting operational losses as a percentage of gross income, of total cost, total budget, or in basis points of capital facilitates the comparability.
- No averages because of its asymmetric distributions
 - Averages hide the heterogeneity. The bias in operational loss averages is often caused by outliers.
 - Better alternatives to averages are the median and the first and third quartiles of the distribution.
- Combined assurance: The three lines of defense should work together to provide combined assurance to the board (RAG).

6 Aggregating qualitative risk data

- The challenge is that risk scores, color ratings, and other indicators are discrete, qualitative, unfit for arithmetic treatment. Solution includes:
 - Conversion and addition
 - Categorization
 - Worst-case reporting
- Conversion and addition: Convert qualitative metrics into monetary unit that is linear, additive arithmetically.
- Categorization: Grouping them by color or score avoids the misleading collapse of heterogeneous information into improper aggregate.
- Worst-case reporting: Conservatism and prudent form, appropriate when risk tolerance is minimal, disadvantage is potentially over-alarming.

7 External risk reporting

- Reporting to the regulator: Pillar 3, Basel regulation
- Reporting to the market and investors: risk section of the annual report
- Reporting on operational resilience and satisfying regulatory expectations

8 External risk reporting-Reporting to regulators

- Operational risk disclosure requirements include three types of information:
 - Qualitative information on operational risk management
 - Historical losses
 - Aggregate operational losses incurred over past 10 years
 - Business indicator and subcomponents
- Absence of evidence is not evidence of absence: Auditor or regulator will not take a risk manager's verbal assertion as confirmation, proof is required.
- Incident reporting: Besides public disclosure, financial institutions are required to notify regulators of any significant operational risk events or any breach of conduct.
 - Torn between need to be compliant with the regulation and reluctance to disclose internal operational risk failures.

9 External risk reporting-Reporting on operational resilience

- In March 2022, regulated financial firms in the UK entered a three-year transitional period. By March 2025, they must:
 - perform mapping and testing so that they can remain within impact tolerances for each important business service.
 - make the necessary investments to enable these services to operate consistently within their established impact tolerances, and report to the regulator on these elements.

考点 7 Integrated Risk Management

1 ERM framework

- Include RMC and four elements: risk governance, risk appetite, risk culture, risk capital and stress testing.
 - Risk governance, culture, and appetite sets the priorities for and guides ERM.

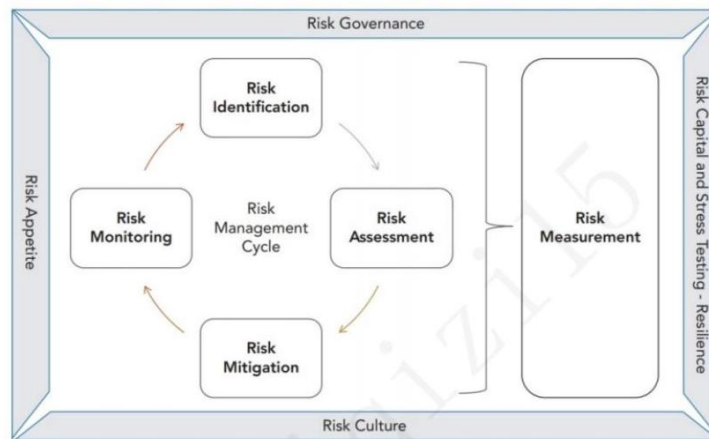


图 8: ERMframework

2 Risk governance

- Risk governance prescribes the roles and responsibilities of individuals in the three lines of defense, and organizes decision-making and reporting, generally through committees.

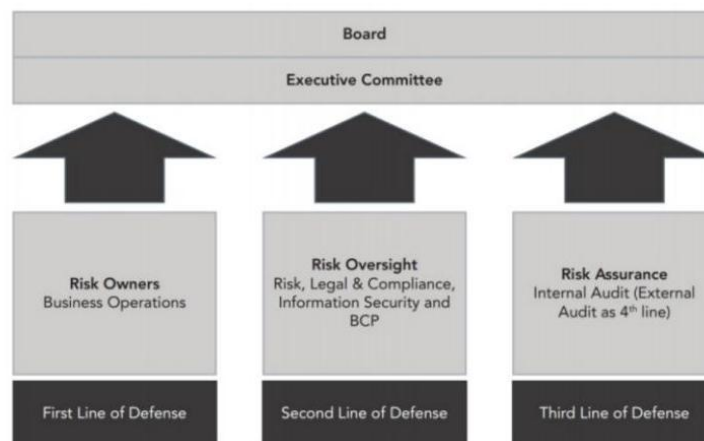


图 9: Riskgovernance

3 Risk culture

- Corporate culture has a direct influence on the attitude and preferences of the firm when managing risks, from prudent to daring, from compliant to challenging.
- Regulators consider risk culture to be a key driver of ERM framework effectiveness.

4 Risk appetite

- At the ERM level, risk appetite is typically expressed in overarching risk appetite statements (or policy).

5 Role of regulatory capital

- The objective of the BCBS prudential regulation is threefold:
 1. To ensure the solvency and soundness of all financial intermediaries
 2. To provide customers protection from undue risks (failure, fraud, opportunistic behavior)
 3. To promote the efficient and competitive performance of financial institutions
- Objective 1 is the primary objective, which is addressed through regulatory capital requirements.
- Objective 2 and 3 are attained through requirements on the bank's senior management's history and competence (the "fit and proper" criteria), monitoring and reporting on the bank's activities.
- However, in the United States, the focus has now shifted from capital estimation to stress testing.
 - Models developed for the Federal Reserve's Comprehensive Capital Analysis and Review (CCAR) program.
 - Stress testing requires institutions to estimate expected losses under specific adverse economic.
 - Capital is a through-the-cycle concept, whereas stress testing is a point-in-time exercise.

6 Role of economic capital

- Financial intermediaries need to calculate economic capital for themselves that fully reflect their risk profile and potential needs to cover unexpected losses.
- The level of economic capital is largely linked to the credit rating, which then influences their cost of funds.
 - The larger the capital, the bigger the buffer against losses, the better the creditworthiness of the institution, the lower the costs of raising funds.

7 Risk aggregation for capital assessment

- The aggregation of the different risk classes is the inter-risk diversification.
 - It supplements the intra-risk diversification, which is the diversification within each risk class.
- Total aggregated capital tends to be lower than the sum of each stand-alone capital amount for individual risk. The difference reflects the diversification benefits.
- Operational risk can bring important diversification benefits to the total capital.
 - Because operational risk typically has a low correlation to other types of financial risks, due to its different risk drivers.

8 Stress testing

- It involves stressing that system or entity beyond normal operation, often to a "breaking point," to observe the results.
 - The ultimate objective is to ensure that institutions can absorb the losses incurred under extreme adverse scenarios and remain a going concern.
 - Stress testing alerts the management to adverse unexpected outcomes and provides an indication of how much capital might be needed to absorb losses.

9 Stress-testing taxonomy-two dimensions

- Dimension 1: Quantitative-Qualitative Approach
 - On the quantitative end of the spectrum, stress-testing approaches relate to the sensitivity of models to parameter shocks.
 - On the qualitative end of the spectrum lie approaches more focused on scenario analysis such as macro stress testing, and non-model-based evaluations like reverse stress testing.
- Dimension 2: Measurable-Immeasurable Risk
 - Range from fact-based probabilistic analysis of measurable risks to more hypothetical possibilistic analysis of immeasurable risk.
 - * On the immeasurable risk end of the spectrum, review and analyze possible risks that are "unknown unknowns". The probability cannot be known, and it cannot be determined which event is more likely than the other.

10 Parameter stress testing

- Parameter (or model) stress testing relates to testing the robustness of a model by changing the value of its parameters.
 - It utilizes quantitative approaches and seeks to analyze measurable risks.
 - Banks will stress model parameters or conduct sensitivity testing to determine the impact on a particular model, portfolio, or the bank.

11 Macroeconomic stress testing

- It takes the most holistic view of risk run bank-wide for all risk types: seek to stress both measurable and immeasurable risks, and ideally to stress the dependency structure.
 - Macroeconomic scenarios relate to inflation, unemployment rate, GDP changes and so forth, and aim at testing the financial resilience of the largest banks.
 - G-SIBs (globally systemically important banks) are provided a list of macro shock scenarios to assess the solvency level.

12 Parameter and macroeconomic stress testing

- Quantitative analysis:
 - Quantitative analysis of parameter stress testing focus on statistical scenarios such as a "standard deviation event" .
 - Quantitative analysis of macroeconomic stress testing focuses on estimating the output of the models based on a set of macroeconomic scenarios.

13 Reverse stress testing

- Reverse stress testing utilizes primarily qualitative approaches and seeks to analyze immeasurable risks.
 - It starts from the identification of a pre-defined outcome and then determines how bad an adverse scenario would need to be to achieve the pre-defined outcome.
- Reverse stress testing is a risk management tool, apt for assessing operational resilience, rather than calculating needed financial resources to weather extreme conditions.

14 Operational risk stress-testing models

- In developing the macroeconomic-based stress-testing models, banks can model either total operational risk losses, or the two components of operational risk losses: frequency and severity (a preferred approach).
 - Severity can be more complex to model than frequency as severity is highly impacted by tail events.
- Once the model produces stressed losses, an expert refinement needs to be performed using scenario analysis.

2 重点风险领域

考点 1 Cyber-Resilience: Range of Practices

1 Framework

- Cyber resilience standards and guidelines
- Cyber governance
- Communication and sharing of information
- Interconnections with third parties

2 Cyber resilience standards and guidelines

- Most jurisdictions address cyber through IT and operational risk standards.
- Even standards focusing on business continuity planning and outsourcing have relevance to cyber-risk.
- A few jurisdictions issued specific cyber-risk management guidance.
 - For example, guidance in Singapore focuses on detecting cyberattacks, while guidance in Brazil focuses on establishing cybersecurity policies.
- Other jurisdictions have no specific cyber-security regulation, international standards/prescriptive guidance are encouraged.

3 Cyber governance

表 4: Five Areas of Cyber Governance

Area	Description
1. Cyber-security strategy	Cyber-security strategy is expected but not required.
2. Management roles and responsibilities	Prioritizes roles and responsibilities of BOD and senior management.
3. Cyber risk awareness culture	Staffs at all levels of a firm should be aware of cyber risk and have cybersecurity awareness. Banks should establish appropriate screening and checks for new employees.
4. Architecture and standard	No common regulatory standards requiring cyber architecture and guidelines are in place. Only a few jurisdictions have specific controls about cybersecurity architecture.
5. Cyber security workforce	On-site inspections (review the training process or qualifications of employees) or questionnaires.

- Cyber-security strategy using three types regulatory approaches:
 - Cyber-security strategy in either sector-specific or across multiple industries, common approach in emerging market.
 - Individual own cyber-security strategies.
 - Examining whether institution has IT strategy and accompanying security provisions, prevalent in Europe.

4 Approaches on cyber-risk management

表 5: Approaches on Cyber-Risk Management

Approach	Description
1. Supervision of cyber-resilience	Survey responses, physical inspections (on-site or off-site), incident reports from auditors or other third parties, in-person meetings.
2. Information security controls	Most jurisdictions recognize the importance of mapping, classifying business services, supporting assets and services as a basis for building resilience. Independent assurance by auditors.
3. Incident response and recovery	Establish a risk framework or policies for risk detection and response and recovery. Training exercises on how to respond to incidents.
4. Cybersecurity and resilience metrics	Forward-looking rather than back-looking indicators help better improve resilience to attacks because adversaries are dynamic.

5 Communication and sharing of information

- The content of the sharing information

- The information include cyber-threat information, information related to cyber-security incidents, regulatory and supervisory responses in case of cyber-security incidents and/ or identifications of cyber-threat, and best practices related to cyber-security risk management.

表 6: Five Types of Cyber-Security Information-Sharing Practices

Type	Description
1. Sharing among banks	(i) No common standard for information-sharing. (ii) Regulators are not directly involved in bank-to-bank information-sharing. (iii) Mainly depend on the financial industry's culture and level of trust.
2. Sharing from banks to regulators	(i) Mandatory requirement and usually limited to cyber incidents. (ii) May have informal flow back.
3. Sharing among regulators	(i) Cyber-fraud is becoming more sophisticated and cross-jurisdiction. (ii) Least frequent flow of information exchange. (iii) Mandatory or voluntary.
4. Sharing from regulators to banks	Regulators are more readily to share publicly available information such as cyber-security risk management best practices.
5. Sharing with security agencies	(i) Create broader awareness of other sectors' cyber threats and enhance effective measures for these cyber threats. (ii) Either voluntary or mandatory.

6 Interconnections with third parties

• Introduction

- All jurisdictions recognize the challenge of gaining assurance of an entity's cyber-resilience, a challenge both for regulators, and for financial institutions' third-party service providers.
- "Third parties" is understood in a broad sense, including:
 - * (i) all forms of outsourcing (including cloud services);
 - * (ii) services and products that are typically not considered outsourcing (power supply, telecommunication lines);
 - * (iii) interconnected counterparties such as FMI.

表 7: Interconnections with Third Parties

Analysis Areas	Description
1. Governance of third-party interconnections	Regulations require institutions to develop board-approved outsourcing framework and contractual framework (roles, responsibilities, and obligations).
2. Business continuity and availability	(i) Recovery and resumption objectives should be defined for critical business activities from an end-to-end perspective. (ii) Prompt transfer outsourced activities to other provider in need. (iii) Test protect measures on a yearly basis and complement by audits and monitoring outsourcing vendors.
3. Information confidentiality and integrity	Explicitly addressed by contractual terms.
4. Specific practices regarding visibility of third-party interconnections	In many jurisdictions, the supervisory requests to be informed about the material outsourcing agreements made by institutions and imposes some conditions on them.
5. Auditing and testing	(i) Regulators require the financial organizations to guarantee “rights to inspect and audit” service providers. (ii) Acquire audit opinion on the outsource arrangements from the external auditor. (iii) Compliance testing: red teaming exercises.
6. Resources and skills	(i) Specialist competencies to assess whether maintaining effective oversight of the new risks posed by new technologies. (ii) Whether have sufficient and qualified personnel to ensure continuity in managing/monitoring outsourced services, even if key personnel leave the institution. (iii) On-site inspections to assess human resources and qualification.

- As regards supervisory practices, following are widespread:
 - Intrusive on-site inspections relating to cyber-risk of outsourcing. The inspections review the outsourcing framework, the applicable processes and the completeness and adequacy of specific risk assessments and contracts.
 - Off-site supervision, jurisdictions receive periodic reports that assess the outsourcing policies and risks.

考点 2 Case study: Cyberthreats and Information Security Risks

1 Information security risks (ISR)

- Taxonomy of IRS using a four-quadrant method:
 - internal causes versus external causes
 - data theft versus data loss

表 8: Information Security Risks (ISR)

Data Incidents	Theft or Corruption	Loss or Unvoluntary Disclosure
External causes or third parties	1. Digital: Hacking, Virus infection, phishing and other cyberattacks 2. Physical: Theft, social engineering	3. Disaster, systems disruptions, third-party failure
Internal causes	4. Theft and transfer of digital or physical information by infiltrated employee or contractor 5. Departing employees take proprietary information or intellectual property from the firm (mishandled exists)	6. Database loss, back-up loss 7. Loss of devices by staff members 8. Errors when sending documents (e-mail recipients or attachments) 9. Loss of printed documents (e.g., by accidentally disposing of them in a wastebasket) 10. Errors or accidental mentions of confidential information when communicating to outsiders 11. Loss of archives

2 Cyber risk management: frameworks

- Three cybersecurity standards dominate the market:
 - US National Institute of Standards and Technology (NIST) Framework for Improving Critical Infrastructure Cybersecurity
 - Center for Internet Security Critical Security Controls (CIS)
 - The International Standards Organization (ISO) frameworks ISO/IEC 27001 and 27002
- The NIST Framework for Improving Critical Infrastructure Cybersecurity
 - The core of the framework is a list of cybersecurity functions that follow the basic steps of cyber defense: identify, protect, detect, respond, and recover.
- Center for Internet Security Critical Security Controls (CIS)
 - CIS was built in the late 2000s by a volunteer-expert coalition to create a framework for protecting companies from the threats of cybersecurity. It is comprised of key controls, regularly updated by experts from all fields.
 - CIS controls are prioritized to mitigate the most prevalent cyberattacks against systems and networks.
- ISO/IEC 27001- International Standard Organization

- The International Standard ISO/IEC 27001: 2013 is the internationally recognized standard for cybersecurity. It provides general guidance to firms to organize the risk management.
- ISO standards do not provide implementable guidance to firms per se, but rather act as evaluation grids for those who want to achieve certification in ISO standards.

3 Essentials of cybersecurity protection and monitoring

- The information security protection has three dimensions referred to as CIA: Confidentiality, Integrity, Availability.
 - The first two relate to information security, while the third relates to business continuity and systems uptime-now in the scope of resilience.
- Information controls can be grouped into two broad categories: Behavioral controls and Technical controls

4 Behavioral controls

- Behavioral controls: address human behaviors and fallibility when it comes to handling and protecting information.
 - The controls include awareness campaigns, rules of conduct and prudence for employees and contractors, online training, password management, supervision, and sanctions.

5 Technical controls

- Technical controls: these relate to all technical aspects of systems, either for prevention or detection.
 - Preventative controls relate to system architecture, access, firewalls, encryption, passwords and patching, and are essentially directed at external threats.
 - Detective controls provide early warnings of data leaks, whether initiated internally or externally.

6 Equifax case study

- On March 10, 2017, hackers breached Equifax's networks by exploiting a vulnerability in one of the systems via Equifax's online dispute portal.
- Attackers quickly moved from the infected portal and gained access to other parts of Equifax's network.
- For three months, hackers remained undetected while accessing multiple Equifax databases and extracting consumers' personal information.

7 Lessons learned from the Equifax case study

- A lack of a comprehensive inventory of IT assets.
- Failure of risk management policy enforcement and failure to enforce the patch management policy.
- Inconsistent communication among employees on the remediation of security vulnerabilities.
- An expired SSL certificate designed to inspect encrypted network traffic.
- Poor external communication as part of crisis management.

考点 3 Sound Management of Risks Related to Money Laundering and Financing of Terrorism

1 Best practices

- Basel Committee is issuing guidelines to describe how banks should include money laundering (ML) and financing of terrorism (FT) risks within their overall risk management.
- The Committee has a long-standing commitment to promote the implementation of sound Anti-Money Laundering and Countering Financing of Terrorism (AML/CFT) policies and procedures that are critical in protecting the safety and soundness of banks and the integrity of the international financial system.
- To implement sound Anti-Money Laundering and Countering Financing of Terrorism (AML/CFT) policies and procedures, banks should not establish a threshold transaction value and review all transactions above the threshold.
- Terrorist screening is not a risk-sensitive due diligence and should be carried out irrespective of the risk profile. A bank should freeze without delay and without prior notice funds or other assets of designated persons and entities.
- Three lines of defense of mitigating ML and FT
 - First line of defense is business units, they must identify, assess and control the risks of their business.
 - Second line of defense is the chief AML/CFT officer who should have direct reporting lines to board. The chief AML/CFT officer should be the contact point regarding all AML/CFT issues for internal and external authorities, including supervisors or financial intelligence units.
 - Third line of defense: Internal and/or external auditors.
- Enhanced due diligence may be required for:
 - An anonymous account with large balance.
 - Large size of cross-border transaction account.
 - An account of foreign politically exposed person (PEP).
- A bank should not open an account with a customer who insists on anonymity (匿名), while confidential numbered accounts can not function as anonymous accounts.
- Even if a customer has opened a bank account at a reputable bank, he can not be categorized as low-risk and well identified. Banks must perform due diligence before performing any transaction.
- While the transfer of funds from an account in the customer's name in another bank subject to the same CDD standard as the initial deposit may provide some comfort, a bank should nevertheless conduct its own due diligence.
- If a bank has any reason to believe that an applicant has been refused by another bank due to concerns over illicit activities, it should consider classifying this applicant as higher-risk and apply enhanced due diligence procedures.
- Automated processes should be employed more frequently to manage ML/FT risk as the bank involves more complicated banking business.
- Regulators encourage rather than discourage financial institutions to apply advanced analytics like machine learning which can help facilitate the detection of anomalies.
- The bank providing correspondent banking services (代理银行服务) to extensively numerous respondent banks tends to increase its ML/FT risk.
- A bank performing business nationally and abroad should appoint a chief AML/CFT officer for the whole group.
- Where the minimum regulatory or legal requirements of the home and host countries differ, offices in host jurisdictions should apply the higher (stricter) standard of the two.
- To effectively manage the ML and FT risks arising from related accounts, a bank should integrate this information based not only on the customer but also on its knowledge of both the beneficial owners of the customer and the funds involved

考点 4 Case Study: Financial Crime and Fraud

1 Financial fraud risk management

- Elements of a control framework to manage internal fraud

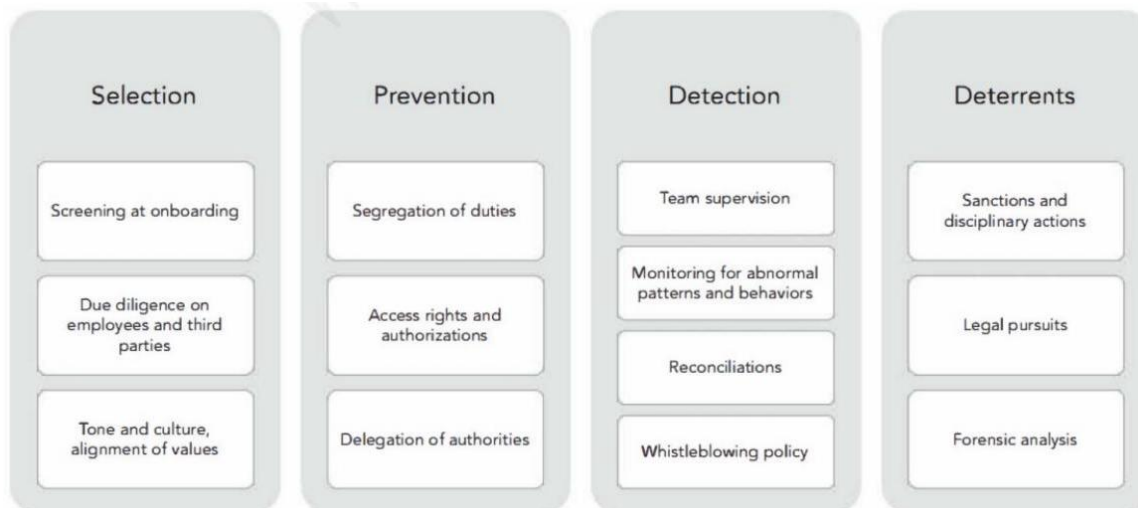


图 10: fraud

2 AML risk management

- The three phases in money laundering are placement, layering, and integration or extraction.
- Elements of a control framework to manage AML risk:

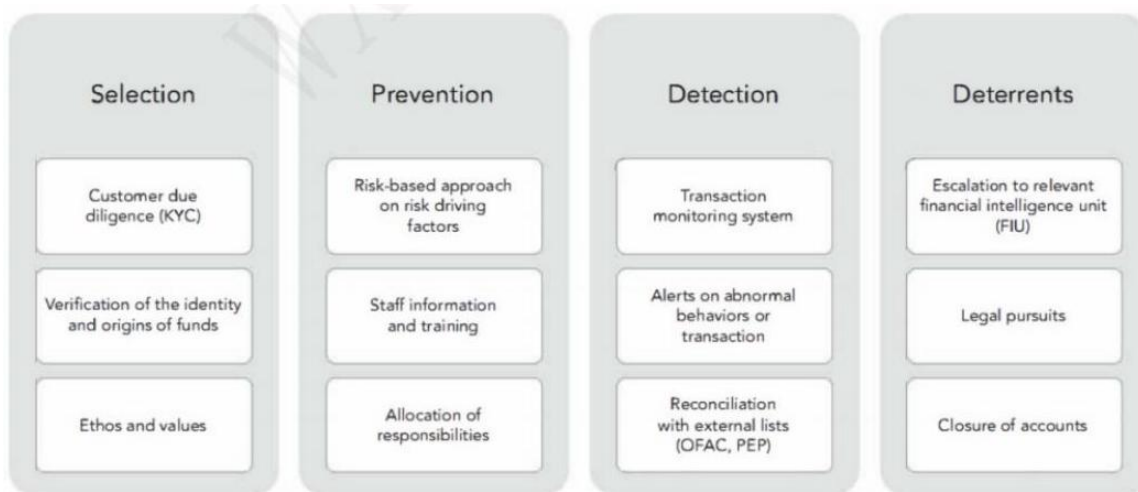


图 11: fraud2

3 Case study: USAA

- USAA paid \$140 million of fines in AML management failure reached with the Financial Crimes Enforcement Network (FinCEN) and the Office of the Comptroller of the Currency (OCC).
- The fines were due to a lack of control over AML risk (weak control environment).

- Regulatory findings and sanctions mean costly AML remediation programs (called lookbacks) in which the bank must review all client files for a specified period, verify client information, file suspicious activity reports and even close suspicious accounts.
- COVID-19 pandemic makes it more difficult for financial institutions to identify anomalies based on automation of detection prone to false positives, false negatives.

考点 5 Guidance on Managing Outsourcing Risk

1 Risks from the use of service providers

- Compliance risks arise when a service provider violates laws/regulations.
- Concentration risks arise outsourced services are provided by limited service providers or in limited geographic locations.
- Reputational risks arise bad performance of a service provider causes the public to have negative opinion on financial institution.
- Country risks arise financial institution engages a foreign service provider, exposing possible risks from the foreign country.
- Operational risks
- Legal risks arise when a service provider exposes a financial institution to legal expenses and possible lawsuits.

2 Elements of effective program to manage outsourcing risk

- Risk assessments
- Due diligence and selection of service providers
- Contract provisions and considerations
- Incentive compensation review
- Oversight and monitoring of service providers
- Business continuity and contingency plans

3 Due diligence of service providers

- Financial institution should conduct an evaluation of and perform the necessary due diligence for a prospective service provider prior to engaging the service provider. The overall due diligence process includes a review of the service provider with regard to three points.
 - Business background, reputation, and strategy
 - Financial performance and condition
 - Operations and internal controls

4 Contract provision

- Terms of service agreements should be defined in written contracts that have been reviewed by the financial institution's legal counsel prior to execution. Elements usually include 15 parts.
 - Scope: contracts clearly define the rights and responsibilities of each party.
 - Cost and compensation: contracts describe the compensation, variable charges, and any fees to be paid for non-recurring items and special requests.

- Right to audit: agreements provide for the right of the institution to audit the service provider/access to audit reports. The provision is optionally included.
- Establishment and monitoring of performance standards: agreements define measurable performance standards for the services or products being provided.
- Confidentiality and security of information: service providers ensure the security and confidentiality of both the institution's information and customer information
- Ownership and license: Agreements should address the ownership of any information generated by service providers. Agreements define circumstances under which service providers may use institution property inclusive of data, hardware, software, and intellectual property.
- Indemnification: agreements provide for service provider indemnification of the institutions for any claims against the institutions resulting from the service provider's negligence.
- Default and termination: agreements define events of a contractual default, list of acceptable remedies, and provide opportunities for curing default. Bad performance and nonperformance of duties detected by performance metrics can lead to default and termination.
- Dispute resolution: dispute resolution process is to expedite problem resolution and minimize disruption.
- Limits on liability: service providers may want to contractually limit their liability.
- Insurance: service providers have adequate insurance and provide financial institutions with proof of insurance.
- Customer complaints: agreements specify the responsibilities of both parties related to responding to customer complaints.
- Business resumption and contingency plan of the service provider: the contingency plan is developed by the service provider and reviewed/tested by the financial institution.
- Foreign-based service providers: the institutions should consider including choice of law and jurisdictional provisions. This would avoid confusing situations where the foreign laws differ substantially from local laws.
 - * Banks may apply different standards for local and foreign-based service providers.
 - * Service agreements with foreign-based vendors should always be flexible to allow the laws of home or host country to apply for resolving related disputes.
- Subcontracting: if agreements allow subcontracting, the same contractual provisions should apply to subcontractor. Service providers rather than financial institutions have overall accountability for all services that the service provider and its subcontractors provide

考点 6 Case Study: Third-Party Risk Management

1 Third-party risk

- Financial firms contract with third parties for services as a cost-savings approach, outsourcing a wide variety of activities to specialized firms with a competitive advantage in particular systems, processes, and knowledge.
- While contracting with third-party providers for back-office activities or cloud-based data management may have the potential to reduce process errors or cybersecurity risks, it does not fully mitigate operational risk and increases exposure to "third party risk".

2 Effective third-party risk management framework

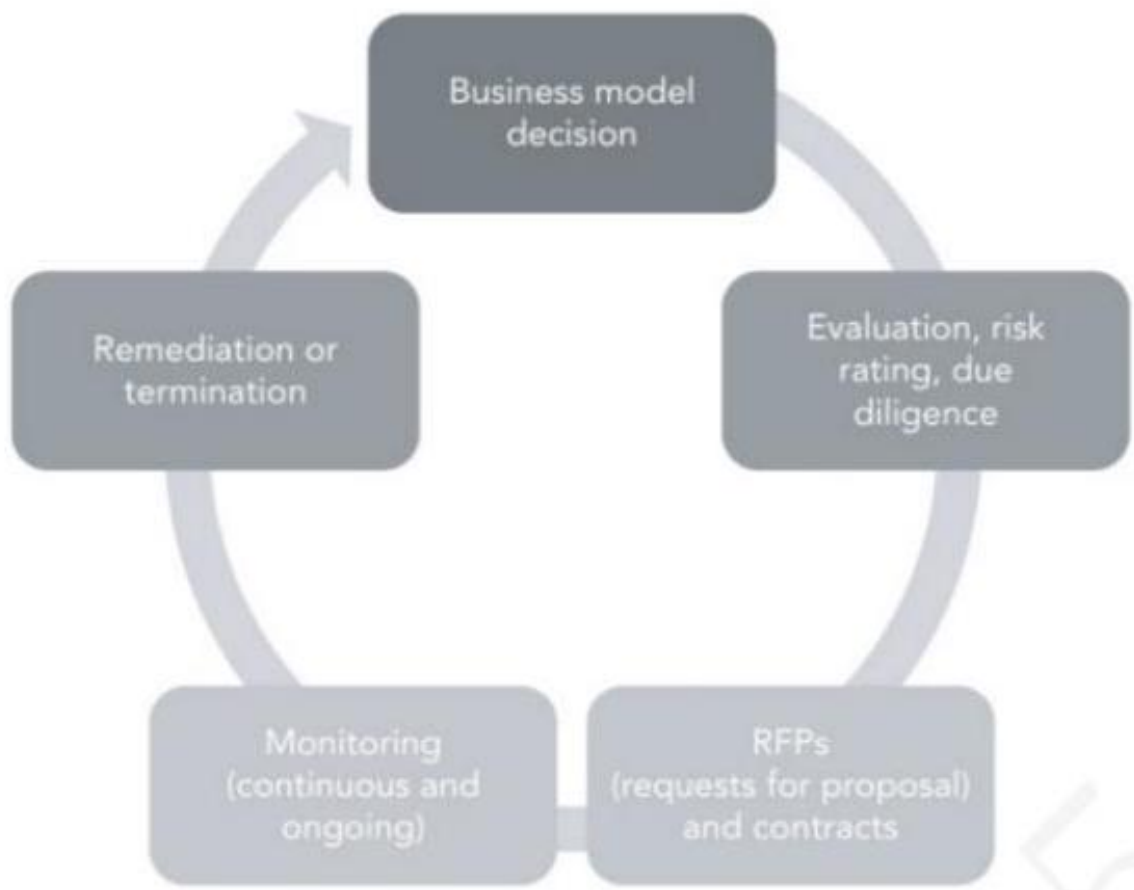


图 12: tpframeworks

3 Case study: Capital One data breach

- Case: Federal authorities ultimately arrested a former Amazon Web Services (AWS) cloud services employee for breaking into the bank's server. The former employee of AWS was charged with stealing 140,000 Social Security numbers and 80,000 bank account numbers in the breach.
- Lessons: The increase in breaches involving cloud databases and services is the result of poor security hygiene

4 Case study: OCC fines Morgan Stanley

- Case: Morgan Stanley received a \$ 60 million fine imposed by the OCC due to failures in the risk management and oversight of vendors in the decommissioning of two wealth management business data servers in 2016.
- Lessons: The bank did not exercise adequate due diligence in selecting the third-party vendors and did not adequately monitor the vendors' performance.

考点 7 Case study:Investor Protectionand Compliance Risks in Investment Activities

1 MIFID (Markets in Financial Instruments Directive)

表 9: Regulations

Regulations	Description
MIFID (2004)	Define regulatory reporting and trade transparency obligation to prevent market abuse and defines the rules on the acceptance of financial instruments for public trading.
MIFID II (2014)	Alongside the reinforcement of investor protection, MIFID II set out additional requirements on public disclosure of data on trading activity, and on disclosure of transaction data to regulators and supervisors.

2 Dodd-Frank

- The Investor Protection Act in the USA in 2009.
 - Whistleblowers are granted increased protections.
 - Mandate derivatives need to be traded in clearinghouses.
 - Aim at ensuring a greater financial stability of the markets.
 - The Volcker Rule intends to prevent commercial banks from engaging in speculative activities and proprietary trading.
 - Create Consumer Financial Protection Bureau (CFPB) .

3 Statistics and record fines cases

- The most severe penalty was imposed on UBS for\$ 11.15 billion in 2008 because UBS misrepresented auction rate securities to investors as safe, cash-equivalent products, when in fact they faced increasing liquidity risk.
- The largest fine for spoofing is held by JP Morgan fined \$ 920 million by the CFTC (Commodity Futures Trading Commission) in September 2020 by US regulators for ”spoofing” in the precious metals and US Treasury markets.

4 FINRA fines Deutsche bank securities

- The Financial Industry Regulatory Authority (FINRA) fines Deutsche bank securities, Inc.\$ 2 million for best execution violations.
- FINRA found that, between 2014 and 2018, Deutsche Bank Securities was routing customer orders to exchanges through its smart order router. This indirect routing of orders created delays of execution in customers’ market orders and also caused lower fill rate.
- Lessons:
 - Regulators tend to apply punitive fines such that the penalty more than offsets the benefits accumulated where firms perpetuate non-compliant practices.
 - The fines are aimed to act as deterrent for further deviations and have a signaling power to peer firms to encourage them to adjust their processes and monitoring practices to ensure compliance.

考点 8 Supervisory Guidance on Model Risk Management

1 Model

- Model refers to a quantitative method, system, or approach that applies statistical, economic, financial, or mathematical theories, techniques, and assumptions to process input data into quantitative estimates.
 - A model consists of three components: an information input component; a process component; a reporting component.
 - Models are simplified representations of real-world, focus on particular aspects considered to be most important.

2 Model risk

- Model risk: the potential for adverse consequences from decisions based on incorrect or misused model outputs/reports.
- Model risk occurs for two reasons:
 - Fundamental errors produce inaccurate outputs against the design objective and intended business uses.
 - The model may be used incorrectly or inappropriately.

3 Effective framework to manage model risk

- Elements of the framework
 - Robust model development, implementation, and use
 - A sound model validation process
 - Governance sets an effective framework with defined roles and responsibilities, as well as the authority to restrict model usage.

4 Elements of the framework — 1.development,implementation

- At initiation, ensure the model is logical and is on the basis of technically-correct mathematical theories.
- Then ensure that the data collected and assumptions are robust and relevant for the model.
- Finally there should be a clear documentation and made clear to users.
- After developing the model:
 - Quantitative aspects: quantitative model testing includes checking the model's accuracy, robust and stable, limitations and the model's assumption/behavior over a range of input values. Testing should be done in both normal and extreme (stressed) market conditions.
 - Qualitative aspects: Banks should ensure that the development of the more judgmental and qualitative aspects of their models is also sound.

5 Elements of the framework-2.validation

- Three core elements of effective validation framework
 - Evaluation of conceptual soundness, including developmental evidence
 - Ongoing monitoring , including process verification and benchmarking
 - Outcomes analysis, including back-testing
- Evaluation of conceptual soundness, including developmental evidence
 - Carry out overall quality check on the model design and construction.
 - * Mainly examine the documentation evidence and perform supplemental sensitivity analysis (one variable change) and testing (multiple variables change) .

- * Qualitative information and judgment in model development should be validated and documented.
- Ongoing monitoring
 - Once the model comes into play, ongoing monitoring begins. Ongoing monitoring tries to determine whether any changes, additional developments, or replacement should be needed for the model.
 - Sensitivity analysis and other checks for robustness and stability should likewise be repeated periodically.
 - Ongoing monitoring should include the analysis of overrides with appropriate documentation.
 - It includes process verification and benchmarking to confirm the model is implemented as intended.
 - * Process verification: Ensure data inputs to be error-free and complete. Execute Internal control to ensure computer code can only be altered by authorized staff.
 - * Benchmarking: Compare a model's outputs with other comparable models or inputs with comparable data. Discrepancies should trigger investigation into the sources and degree of the differences.
- Outcomes analysis, including back-testing
 - Rely on quantitative approaches such as statistical tests and qualitative approaches such as expert judgment.
 - Outcomes analysis should be conducted on an ongoing basis. Confidence intervals need to be developed based on the model estimates, so that outcomes that are beyond the intervals should be specially focused on and monitored.
 - Parallel outcomes analysis is based on that models are frequently revised for new information or data.

6 Challenges to an effective validation process

- System integration can be a challenge because the model need draw from various sources of data to process huge data, and feeds into multiple data repositories and reporting systems.
- The widespread use of vendor/third-party products poses unique challenge because the modeling expertise is external to the user and some components may considered proprietary.

考点 9 Case Study: Model Risk and Model Validation

1 Model risk

- There are two broad risks an institution can be exposed to due to model risk.
 - Execution risk: occurs when a model does not do what it is intended and designed to do.
 - Conceptual errors: occur when the wrong assumptions or modeling techniques are used in the model.

2 Model risk management (MRM) function

- MRM function sits in the second line of defense and is staffed with experts who are independent of model development.
- To balance the cost and burden of model validation with the need to ensure that model risk is adequately addressed, MRM functions assign models to different tiers according to the risk they pose to the institution.
- Highest-tier models will be reviewed in detail, with the validation team performing comprehensive back-testing and assessing the model.
- High-tier models also undergo more frequent full scope validations, usually every 2 to 3 years.
- Lower-tier models are subject to a similar process but with the depth and frequency of the various actions tailored to the tier.
- All models, independently of their tier, undergo an annual review of the environment.
- Model risk management should be seen as a continuous effort rather than a point-in-time exercise revolving around validations and reviews.

3 Gaussian Copula and CDO pricing

- Case: Correlation assumption is a problem. Banks fail to update the copula model with the new correlation estimates implied by new CDS prices, and continue to price CDOs using the old assumptions.
- Lessons: The critical responsibility of MRM function is to challenge the assumptions upon which the model relies and ensure that users understand its limitations.

4 Barclays' acquisition and excel spreadsheet error

- Case: Barclays Capital offered to acquire Lehman Brothers' positions after Lehman Brothers filed for bankruptcy, including 179 trading contracts that Barclays did not want to buy. They were hidden rather than deleted. A junior law associate reformatting the Excel file to a PDF makes hidden rows visible again in the PDF file.
- Lessons: Seemingly simplistic tools or models still need the right review, challenge and controls.

5 NASA Mars Orbiter

- Case: NASA lost a \$125 million Mars orbiter because a Lockheed Martin engineering team used English units of measurement, while the agency's team used the more conventional metric system.
- Lessons: This is an error in the "assumptions" of the model, the choice of the unit (metric vs imperial). A small subset of these mistakes can result in catastrophic losses (\$125 million).

3 Capital Planning

考点 1 Risk Capital Attribution and Risk-Adjusted Performance Measurement

1 Risk capital, economic capital, & regulatory capital

- Risk capital is the cushion that provides protection against the various risks (i.e., unexpected loss) inherent in the business of a corporation so that the firm can maintain its financial integrity and remain a going concern even in the event of a near-catastrophic worst-case scenario.
- Risk capital is often called economic capital, and in most instances the generally accepted convention is that risk capital and economic capital are identical.
- Regulatory capital
 - First, regulatory capital applies to regulated industries to protect the interests of small depositors or policy holders.
 - Second, regulatory capital is calculated by regulatory rules and formulas and sets a minimum required level of capital adequacy.
 - Third, even if regulatory and risk capital are similar numbers, they may not be similar for each constituent business line.

2 Uses of economic capital-traditional way

- Economic capital is really important to financial institutions.
 - Capital is primarily used in a financial institution not only to provide funding but also to absorb risk.
 - A bank's target solvency is a vital part of the product the bank is selling.
 - Although bank creditworthiness is critical, banks are also highly opaque institutions.
 - Banks operate in highly competitive financial markets.

3 Uses of economic capital-new way

- Economic capital's new uses including:
 - Performance measurement and incentive compensation at the firm, business unit, and individual levels
 - Active portfolio management for entry/exit decisions
 - Pricing transactions: calculate risk-based pricing for individual transactions. Each transaction considers the expected loss and the cost of economic capital allocated.

4 RAROC: risk-adjusted return on capital

- $$\text{RAROC} = \frac{\text{after-tax expected risk-adjusted net income}}{\text{economic capital}} \quad (1)$$
 - After-tax expected risk-adjusted net income = expected revenues - costs - expected losses - taxes + return on risk capital +/- transfers
 - Expected revenues are the revenues that the activity is expected to generate (assuming no losses).
 - Costs are the direct expenses associated with running the activity.
 - Expected losses are also called PD amount, they correspond to the loan loss reserve (or provision) that the bank must set aside as the cost of doing business.
 - Taxes are the expected amount of taxes imputed to the activity using the effective tax rate of the company.
 - Return on risk capital is the return on the risk capital allocated to the activity. It is generally assumed that this risk capital is invested in risk-free securities.
 - Transfers correspond to transfer pricing mechanisms, primarily between the business unit and the treasury group, such as charging the business unit for any funding cost incurred by its activities and any hedging cost.
- Economic capital is sum of risk capital and strategic capital
 - Economic capital = risk capital + strategic capital
 - Strategic capital = goodwill + burned-out capital
 - Economic capital = risk capital + goodwill + burned-out capital
 - Strategic risk capital refers to the risk of significant investments about whose success and profitability there is high uncertainty.
 - Burned-out capital refers to the idea that capital is spent on the initial stages of starting up business, but the business may ultimately fail.
 - Goodwill is the amount paid above the replacement value of the net assets (assets - liabilities) when acquiring a firm.

5 Example

- Claire bank has a 1 billion loans with 20% return rate, the direct cost of this loan is 10%, the expected loss is 5%, and the return on government security is 2%. This firm has 500 million economic capital. This firm has 25% effective tax rate, compute the RAROC for the firm.

6 Answer

- Expected revenue = 20% X 1,000 = 200 million
- Direct cost = 10% X 1,000 = 100 million
- Expected losses = 5% X 1,000 = 50 million

- Return on risk capital = 2% X 500 = 10 million
- Transfers = 0
- Economic capital = 500 million
-

$$RAROC = \frac{(200 - 100 - 50 + 10) \times (1 - 25\%)}{500} = \frac{45}{500} = 9\% \quad (2)$$

7 RAROC calculation issues

- Time horizon choice for RAROC is somewhat subjective.
- PD is used for economical capital calculation on an anticipatory or ex ante basis.
 - A point-in-time (PIT) PD is reasonable for near-term expected losses and individual financial instrument.
 - A through-the-cycle (TTC) PD is more reasonable for economic capital, current profitability, and strategic decisions.
 - PIT is more volatile than TTC.
- Confidence level: setting a lower confidence level may reduce the amount of risk capital allocated to an activity.
 - Lower confidence level → Lower economic capital → Higher RAROC → Affect capital allocation decisions (allocating risk capital to business units and individual transactions for the purpose of measuring economic performance)
 - * Once capital has been allocated, each business unit manages its risk so that it does not exceed its allocated capital.
- The confidence level in economic capital calculation should be consistent with firm's target credit rating and can be regarded as a quantitative expression of the risk appetite of the firm.
 - Business units tend to apply lower confidence levels than the firm-wide target confidence level.

8 RAROC for capital budgeting decision

- Hurdle rate (h_{AT}) is after-tax weighted average cost of equity capital to compare with RAROC to make investment decision.

$$h_{AT} = \frac{CE \times R_{CE} + PE \times R_{PE}}{CE + PE} \quad (3)$$

CE/R_{CE} and PE/R_{PE} is the value/cost of common equity and preferred equity, respectively.

- If RAROC > hurdle rate, add value to the firm, the project should be accepted.
- If RAROC < hurdle rate, destroy value for the firm, the project should be rejected or closed down.
- ARAROC: Adjusted RAROC

$$ARAROC = RAROC - \beta_{CE} (\bar{R}_M - r_f) \quad (4)$$

$\bar{R}_M$ is the expected return on the market portfolio, β_{CE} is the beta of the equity of the firm.

 - ARAROC > r_t : Accept projects
 - ARAROC < r_t : Reject projects
- Risk capital and diversification: risk capital for the firm should be significantly less than the sum of the stand-alone risk capital of the individual business units due to diversification.
 - Stand-alone capital is the capital used by mutual independent view among business units.
 - Fully diversified capital is the capital attributed to each activity X and Y, considering diversification from combining.
 - Marginal capital is the additional capital required by an incremental business, considering diversification.

9 Example

- Assume a business unit in a bank plans to carry out only two activities, X and Y:
- Activity X alone requires \$40 of risk capital.
- Activity Y alone requires \$50 of risk capital.
- Activities X and Y combined need a total of \$70 of risk capital.

10 Answer

表 10: Total Diversification Benefit

Total diversification benefit	X	Y
Stand-alone capital	/	40
Fully-diversified capital	20	$40 - 20 \times \frac{40}{90} = 31.11$
Marginal capital	/	$70 - 50 = 20$
	$70 - 40 = 30$	

考点 2 Range of Practices and Issues in Economic Capital Frameworks

1 Overview of economical capital framework

- Economic capital has broadened to combine the underlying risks (building blocks) into an overall aggregation. The building blocks:
 - Use of economic capital and governance
 - Risk measures
 - Risk aggregation
 - Validation of economic capital
 - Dependency modelling in credit risk
 - Counterparty credit risk
 - Interest rate risk in the banking book

2 Issues in building blocks: governance

- Senior management need to provide credible commitment, recognize the importance of using economic capital measures, put adequate resources and fully understand the limitations of economic capital measures.

3 Issues in building blocks: risk measure

- Banks use a variety of risk measures such as VaR and ES.
- All risk measures have advantages and disadvantages which need to be understood within the context of their intended application.

4 Issues in building blocks: risk aggregation

- Practices and techniques in risk aggregation are generally less sophisticated than the methodologies that are used in measuring individual risk components.
- Since individual risk components are typically estimated without much regard to the interactions between risks (e.g., between market and credit risk), the aggregation methodologies used may underestimate overall risk even if "no diversification" assumptions are used.
- There are five commonly used aggregation methodologies.
 - Simple summation: assumes equal weighting, does not take into account the diversification benefits ($p = 1$).
 - Constant diversification: subtracts a fixed diversification percentage from the overall amount.
 - Variance-covariance matrix: Estimates of inter-risk correlations are difficult and costly to obtain. It does not adequately capture nonlinearities and skewed distributions, leading to underestimate the risk.
 - Copulas: combines marginal probability distributions into a joint probability distribution.
 - * It offer greater flexibility in the aggregation of risks and promise a better approximation of the true risk distribution.
 - * More demanding input and building a joint distribution is very difficult.
 - Full modeling/simulation: simulates the impact of common risk drivers on all risk types and construct the joint distribution of losses.
 - * Theoretically, the most appealing method, potentially the most accurate method.
 - * High information technology demands, the process is time consuming, and it may provide a false sense of security.

5 Issues in building blocks: validation

- The validation of economic capital models is at a very preliminary stage. There exists a wide range of validation techniques, each of which provides evidence for (or against) only some of the desirable properties of a model.

6 Issues in building blocks: dependency modeling

- A particularly important and difficult aspect of portfolio credit the dependency structure, both linear and non-linear relationships between obligors.
- The main concern in the area continues to center on the accuracy and stability of correlation during times of stress.
- The correlation estimates provided by current models still depend heavily on explicit or implicit model assumptions.

7 Issues in building blocks: counterparty credit risk

- Measurement of counterparty credit risk represents a complex exercise, as it involves gathering data from multiple systems, millions of transactions, time horizons ranging from overnight to thirty or more years, tracking collateral and netting arrangements, thousands of counterparties.
- This complexity creates unique market-risk-related challenges and operational risk-related challenges.

8 Issues in building blocks: interest rate risk

- The main challenges for interest rate risk in the banking book is the long holding period for balance sheet and modeling indeterminate cash flows due to embedded optionality in many banking book items.
- If not adequately measured and managed, the asymmetrical payoff characteristics of instruments with embedded option features can present risks that are significantly greater than the risk measures suggest.

9 Recommendations

- Based on BIS recommendations, there are 10 aspects:
 - 1. Use economic capital model in assessing capital adequacy
 - * A bank demonstrates how economic capital model is integrated into business decision-making process and understanding of gap between gross (standalone) and net (diversified) risk.
 - 2. Senior mgt: credit commitment to ensure meaningfulness / integrity of process and adequate resources.
 - 3. Transparency and integration into decision making
 - 4. Risk identification: not all risks can be directly quantified.
 - 5. Risk measures: no singularly preferred risk measure.
 - 6. Risk aggregation: depends on quality of the measure of individual risk components and interactions between risks.
 - 7. Validation: Validation should ensure the capital generated by the model is sufficient and model is fit for purpose.
 - 8. Dependency modeling in credit risk: using adequate supplementary approaches (sensitivity analysis, scenario analysis) to address the limitation of credit models.
 - 9. Counterparty credit risk: complementary measurement processes such as stress testing should be used.
 - 10. Interest rate risk in banking book: close attention to the instruments with embedded option, which can present significant risk.

10 Benefits and impacts of EC using and governance

- In order to achieve a common measure across all risks and businesses, economic capital is often parameterized as an amount of capital that a bank needs to absorb unexpected losses over a certain time horizon at a given confidence level.
- Because expected losses are accounted for in the pricing of a bank's products and loan loss provisioning, it is only unexpected losses that require economic capital.

11 Benefits and impacts of EC using

- Credit portfolio management
 - Benefit
 - * Economic capital is a measurement of the level of concentration. A credit's incremental risk contribution is used to determine the concentration of the credit portfolio.
 - * The bank allocates economic capital according to the credit asset's marginal contribution to the bank's total portfolio risk rather than its standalone risk.
 - Impact
 - * Credit portfolio management allows banks to specify appropriate hedging strategies to reduce portfolio concentration, protecting against risk deterioration
- Risk-based pricing
 - Benefit
 - * In theory, under the assumption of competitive markets, prices are exogenous to banks acting as price-takers.
 - * Decisions are driven by a floor/threshold (the minimum RAROC) computed according to the amount of economic capital allocated to the deal.
 - Impact
 - * Exceptions (specific customer relationship/reputational side-effect) to the rule are strictly monitored and require the decision be elevated to a higher level of management.

- Customer and product profitability analysis
 - Benefit
 - * The measurement of performance can be extended down to the customer level, through the analysis of customer profitability to provide the relative riskadjusted profitability of customer relationships and products.
 - Impact
 - * Segmenting customers and resources re-allocating based on more efficient profitable relationships.
- Management incentives
 - Benefit
 - * Economic capital could affect the objective functions of decision-makers at the business unit level and be used in the incentive structure for business-unit management.
 - Impact
 - * Incentives are the most sensitive element for a majority of bank managers and motive them involved in the technical aspects of the economic capital allocation process.
 - * RAROC can motivate business unit managers by enhancing their incentive compensation as less economic capital is allocated to their respective business unit.

考点 3 Capital Planning at large Bank Holding Companies

1 Overview of Fed capital plan rule

- Federal Reserve's Capital Plan Rule requires all U.S.BHCs to have a capital plan by a robust capital adequacy process. The capital plan covers all U.S. domiciled BHCs with total consolidated assets equal to \$50 billion or more.
- Principle 1: Sound foundational risk management
- Principle 2: Effective loss-estimation methodologies
- Principle 3: Solid resource-estimation methodologies
- Principle 4: Sufficient capital adequacy impact assessment
- Principle 5: Comprehensive capital policy and capital planning
- Principle 6: Robust internal controls
- Principle 7: Effective governance

2 Foundational risk management

- The BHC has a sound risk-measurement and riskmanagement infrastructure to support identification, measurement, assessment and control of all material risks.
 - Stronger risk-identification practices include standardized processes (regularly update, review risk exposures).
 - Lagging practice: did not try to account for reputational risk, strategic risk and compliance risk.

3 Loss-estimation methodologies

- The BHC has processes for translating risk measures into estimates of losses over a range of stressful scenarios and for aggregating those estimated losses across the BHC.
 - Apply conservative approaches to estimate losses.
 - Stronger scenario-design practices have clear narrative describing the vulnerabilities/material risks the firm facing.

- Stronger scenario-design practices used internal models in combination with expert judgment rather than relying solely on either.
- BHCs can use third-party scenarios with stronger practices tailoring third party-defined scenarios to their own risk profiles and unique vulnerabilities.
- Leading practice: In evaluating modeling program of BHC, BHC should strive to estimate loss models at granular level.
- BHCs should use internal operational-loss data as a starting point to provide historical perspective, and then incorporate forward-looking elements, idiosyncratic risks, and tail events to estimate losses.
- Lagging practice: model the loss given default of the bank's loan portfolio using a weighted-average approach at the aggregate portfolio level. Leading practice clearly tied LGD to underlying risk drivers.
- BHCs with leading practices used automated processes, while BHCs with lagging practices exhibited a high degree of manual intervention in the aggregation process.
- By design, estimates based on long-run average behavior, including periods of economic expansion and downturn, are not appropriate for projecting losses under stress.
- It is weak to combine two models: roll-rate model and net charge-off model.

4 Resource-estimation methodologies

- The BHC has a clear definition of available capital resources.
 - Quantitative basis and qualitative projections (expert judgment);
 - Conservatism and credibility;
 - Documentation.

5 Capital policy and capital planning

- BHC has capital policy for establishing capital goals, capital levels and composition of capital, making decisions about capital actions, and setting capital contingency plans.
 - BHC should establish capital targets above capital goals.
 - Contingency actions should be flexible enough and realistic for what is achievable during periods of stress.
 - Lagging practice: establish triggers based on actual results but not on projected results, or based on minimum regulatory capital ratios.

6 Assessing capital adequacy impact

- The BHC has processes for estimating of losses and capital resources to assess impacts on capital adequacy.
 - BHCs should have a process for projections of the size and composition of on/off-balance sheet and RWA.
 - Balance projections are a key input to enterprise-wide scenario analysis given direct impact on the estimation of losses, PPNR (pre-provision net revenue), and RWA.
 - Use a full-revaluation approach to model the bank's nonlinear positions (such as complex derivative exposures).

7 Internal controls

- The BHC has robust internal controls governing capital adequacy process components, including policies and procedures; change control; model validation and independent review; comprehensive documentation; and review by internal audit.
 - BHCs should ensure that validation covers all models and assumptions used for capital planning purposes, including any adjustments management has made to the model estimates (management overlay).

8 Governance

- The BHC has effective board and senior management oversight of the capital adequacy plan (CAP).
 - Board has final oversight accountability for capital planning and should receive related information at least quarterly.
 - Senior management is responsible for ensuring that capital planning activities authorized by the board are implemented in a satisfactory manner and is accountable to the board for the effectiveness of those activities.

考点 4 Stress Testing Banks

1 Stress testing - historical evolution

- Supervisory capital assessment program (SCAP) is a US bank stress test in 2009.
- A successful macro-prudential stress testing program, also the first macro-prudential stress test after the 2007-2008 financial crisis.
 - Macro-prudential focuses on the soundness of the banking system as a whole (systematic risks), but micro-prudential focuses on the soundness of the individual institution.
- The 2009 SCAP tested macro scenarios with three main variables: growth in GDP, unemployment rate, and the house price index (HPI).
- SCAP has at least two components:
 - Assessment of institutions: capital strength vs. capital "hole" that needs to be filled
 - A credible way of filling that hole
- Features of stress testing, pre- and post-SCAP

表 11: Comparison of Pre-SCAP and Post-SCAP

Pre-SCAP	Post-SCAP
Mostly single shock	Broad macro scenario and market stress
Product or business unit level	Comprehensive, firm-wide
Static	Dynamic and path dependent
Not usually tied to capital adequacy	Explicit post-stress common equity threshold
Losses only	Losses, revenues and costs

- Before 2011 all supervisory stress tests applied the same scenarios on all banks (i.e., one-size-fits-all approach).
- Mechanically uniform approaches pose problems, so the 2011 and 2012 CCAR (Comprehensive capital analysis and review) asked banks to submit results from banks' own stress scenarios (baseline and stress) besides the supervisory stress scenario in order to bring banks' own vulnerabilities to light.
- 2011 CCAR required that only the macro scenario was published (disclosed), but no bank level results.
- 2012 CCAR disclosed nearly the same level of detail as the 2009 SCAP, namely bank-level loss rates and dollar losses by major regulatory asset class.
 - Disclosure was an important trait of the 2009 SCAP.
 - * Before 2009 SCAP: disclosed realized losses
 - * After 2009 SCAP: disclosed projected losses
- 2011 European banking authority (EBA) conducts European stress test.

- The EBA test includes both a baseline scenario and a stress scenano.
 - The EBA also required significant disclosures. The disclosure was more extensive, including information on exposure by asset class and by geography. All bank level results were available to download in spreadsheet form.

2 Stress testing-challenges in designing scenarios

- Principal challenges in designing stress scenarios is coherence, especially inherently multi-risk factors, making it more difficult to design a coherent stress test.
 - The supervisor's key challenge is to state the joint outcomes of inherently multi-risk factors.
 - Not every thing becomes terrible simultaneously.
 - * Currency depreciation vs. currency appreciation, riskier asset vs. safe haven assets (flight to quality).

3 Stress testing-challenge in modeling losses and revenues

- From macro-scenario to micro-outcome: losses/revenues, the first task is to map from the few macro-factors into the many intermediate risk factors that drive losses.
- In a stress model, the starting balance sheet generates first quarter's income and loss, which in turn determines quarterend balance sheet. The bank must maintain its capital and liquidity ratios level during all quarters of the stress test time horizon, typically for two years.

4 巴塞尔协议

考点 1 Capital Regulation Before the Global Financial Crisis

1 Basel I

- First put forth in 1988, Basel I was fully implemented by the end of 1992.
- It includes:
 - Credit risk
 - Capital requirements

2 Credit risk-standardized approach

- Risk-based capital ratio introduced risk-weighted assets(RWA). It included three categories:
 - Those arising from on-balance sheet assets (excluding derivatives)
 - Those arising for off-balance sheet items (excluding derivatives)
 - Those arising from derivatives
- RWA: risk weights for on-balance-sheet items

$$\sum_{i=1}^N w_i L_i \quad (5)$$

- w_i is the risk weight, and L_i is the principal amount.

• Example

Niu and Son's Bank holds \$100 million of Oman's debt (lessdeveloped country), \$100 million of corporate bonds, \$50 million of uninsured residential mortgage loans, \$50 million of gold bullion. Calculate the total of the risk weighted assets.

表 12: Risk Weights and Asset Categories

Risk Weight (%)	Asset Category
0	Cash, gold bullion, claims on OECD governments such as Treasury bonds or insured residential mortgages
20	Claims on OECD banks and OECD public sector entities such as securities issued by U.S. government agencies or
50	Uninsured residential mortgage loans
100	All other claims such as corporate bonds and less-developed country debt, claims on non-OECD

- Answer
 $100 \times 1.0 + 100 \times 1.0 + 50 \times 0.5 + 50 \times 0 = 225$ million
- RWA: risk weights for off-balance sheet items.
 A credit equivalent amount (CEA) is calculated by applying a conversion factor to the principal amount of the instrument. For example, bankers' acceptances and guarantees are offbalance sheet items.
 - Bankers' acceptances have a conversion factor of 100
- RWA: derivatives under current exposure method
- The credit equivalent amount is $\max(V, 0) + D$ (add-on amount).
 - V is the current market value of the derivative, D is determined by a(add-on factor) and L(the principal amount). D accounts for changes of future market value.

表 13: Risk Weights by Remaining Maturity

Remaining Maturity (yr)	Interest rate	Exchange rate and gold	Equity	Precious metals except gold	Other commodities
<1	0.0%	1.0%	6.0%	7.0%	10.0%
1 to 5	0.5%	5.0%	8.0%	7.0%	12.0%
>5	1.5%	7.5%	10.0%	8.0%	15.0%

- Example
 Niu and Son's Bank has entered into a \$100 million orange juice forward contract with remaining life of 1.5 years. Current value is 8 million and add-on factor is 12%. Calculate the credit equivalent amount.
- Answer
 $8 + 12\% \times 100 = 20$ million
- RWA: derivatives under original exposure method (only for interest rate and foreign exchange contracts)
- The credit equivalent amount is D (add-on amount).
 - D is determined by a(add-on factor either in original maturity or remaining maturity) and L(the principal amount).
- Total risk-weighted assets: calculated as following:

$$\sum_{i=1}^N w_i L_i + \sum_{j=1}^M w_j^* C_j \quad (6)$$

- L_i is the principal of the i_{th} on-balance-sheet asset and w_i is the risk weight for the asset.
- C_j is the credit equivalent amount for the j_{th} derivative or other off-balance sheet item and w_j^* is the risk weight of the counterparty for this j_{th} item.

3 Capital requirement

- Basel I required capital at least 8% of the total RWA. It includes:
 - Tier 1 Capital (core capital for solvency) at least 4% of RWA
 - * Common Equity and disclosed reserves minus goodwill
 - * Noncumulative perpetual preferred stock
 - Tier 2 Capital (recapitalization of entity in resolution/reduce the impact of failures on depositors)
 - * Loan loss reserves not allocated to impairment of certain assets
 - * Undisclosed reserves
 - * Hybrid instruments

4 The 1995 and 1996 Amendments

- Continuous efforts on unsolved issues pushed the updates of 1995 and 1996 amendments.
It includes:
 - Credit risk-adjustment for derivatives
 - Market risk
 - Capital requirements
 - Others

5 Credit risk-adjustment for derivatives

- Netting: participants in the over-the-counter derivatives market began to sign an International Swaps and Derivatives Association (ISDA) master agreement.
- Netting is a clause in ISDA master agreement, stating in the event of a default, all transactions are considered as a single transaction.
 - If a company defaults on one transaction, it must default on all transactions.
- Contrast a financial institution has a portfolio of N derivatives with a particular counterparty and that the current value of the i_{th} derivative V_i .
 - Without netting, the financial institution's exposure in the event of a default today is $\sum_{i=1}^N \max(V_i, 0)$
 - With netting, it is $\max\left(\sum_{i=1}^N V_i, 0\right)$
Without netting, the exposure is payoff from a portfolio of options.
With netting, the exposure is payoff from an option on portfolio.
- Net replacement ratio (NRR), is the ratio of the current exposure with netting to the current exposure without netting.

$$NRR = \frac{\max\left(\sum_{i=1}^N V_i, 0\right)}{\sum_{i=1}^N \max(V_i, 0)} \quad (7)$$

- The credit equivalent amount (CEA) was modified to

$$\max\left(\sum_{i=1}^N V_i, 0\right) + (0.4 + 0.6 \times NRR) \sum_{i=1}^N D_i \quad (8)$$

- Example
 - Consider following portfolio of three derivatives that a bank has with a particular counterparty with the previous table on add-on amount.

表 14: Transaction Details

Transaction	Principal, L	Current value, V	Add-on amount, $a \cdot L_i$
3-year interest rate swap	1000	-60	5
6-year foreign exchange forward	1000	70	75
9-month option on a stock	500	55	30

- Answer

- The current exposure with netting is $-60 + 70 + 55 = 65$. The current exposure without netting is $0 + 70 + 55 = 125$. Net replacement ratio is $65/125 = 0.52$. The total of the add-on amounts, $\sum_{i=1}^N a_i L_i$ is $5 + 75 + 30 = 110$.
- The credit equivalent amount when netting is $65 + (0.4 + 0.6 \times 0.52) \times 110 = 143.32$. Without netting, the credit equivalent amount is $125 + 110 = 235$.
- If the risk weight is 0.2, the risk-weighted assets with netting is $0.2 \times 143.32 = 28.66$. Without netting, it is $0.2 \times 235 = 47$.

6 Market risk

- Market risk capital charge for trading book

表 15: Comparison of Trading Book and Banking Book

	Trading book	Banking book
Purpose	Held for trading	Held to maturity
Time horizon	Short	Long
Risk type	Market risk	Credit risk
Confidence level	99% VaR	99.9% VaR
Example	marketable securities, most derivatives held by the bank for trading	Bank loans, some debt securities

- It details the separate treatment of general market risk and specific risk
 - MR (general market risk): general movements in market prices(such as interest rate risk).
 - SR (specific risk): driven by idiosyncratic changes in a specific position's value(such as changes in a firm's credit spread or a firm's stock price).
- Corporate bonds are exposed to interest rate risk, captured by MR, and credit risk, captured by the SR.
- Two approaches to calculate market risk capital
 - Standardized approach
 - Internal model-based approach

7 Market risk-standardized approach

- The approach assigns capital separately to each item. It ignores correlations between items and assumes perfectly correlated.
 - It details separately for five categories of positions
 - * Fixed income securities and interest rate derivatives except options;
 - * Equity securities and equity derivatives other than options;
 - * Foreign exchange;

- * Commodities;
- * All types of options

8 Market risk-internal model-based approach

- Internal model-based approach embodied a major change in philosophy by permitting banks to use internally developed risk measures as the inputs to formulas specified by regulators.
- Total capital for trading assets= 0.08*12.S* total (MR+ SR)

$$MR = \max (VaR_{t-1}, m_c \times VaR_{avg}) \quad (9)$$

- VaR calculated with 10-day time horizon/99% confidence level.
 - * It means a 1% chance of being exceeded over a 10-day period, m_c is a multiplicative factor with the minimal value of 3.

9 Market risk-specific risk

- Total capital for trading assets= 0.08*12.S* total (MR+ SR)
 - Capital for specific risk, which was required for fixed income, equity instruments, and derivatives, could be determined using either the standardized approach or the bank's internal models.
 - * In using internal models, banks calculate a 10-day 99% VaR for SR. The multiplier was 4 rather than 3 and capital for specific risk could not be less than half of capital calculated using the standardized approach.

10 Capital requirements

- Tier 3 capital
 - Composed mainly of unsecured subordinated debt with an original maturity of at least 2 years, that could be used to meet part of market risk capital requirement.

11 Others

- Back-testing
 - Banks are required to daily back-testing one-day, 99%VaR over the previous 250 days.
 - Number of exception in this back-testing determines m_c .
 - * Less than 5 (0-4 days) , m_c equals 3.
 - * 5, 6, 7, 8, or 9, m_c equals 3.4, 3.5, 3.65, 3.75, and 3.85, respectively.
 - * More than 9 (10 or more days) , m_c equals 4.

12 BASEL II

- In 1999, the Basel Committee proposed new rules which was known as Basel II. The final revision was in 2004. It includes:
 - Three pillars
 - Credit risk
 - Operation risk

13 Three pillars by BCBS

- Basel II developed three pillars: minimum capital requirements, supervisory review process, and market discipline.
 - Regulatory capital for Pillar 1 determines the minimum level of capital that banks need to cover the risk of unexpected losses from credit, market, and operational risks, and the minimum ratios required.
 - Supervisory review process: Pillar 2 includes additional capital requirements ("add-ons") to Pillar 1 capital for the specific risk profile of a regulated entity.
 - Market discipline: Disclose qualitative and quantitative information.

14 Credit risk

- For credit risk, Basel II specified three approaches:
 - Standardized approach
 - Foundation internal ratings based (IRB) approach
 - Advanced IRB approach

15 Credit risk-standardized approach

- It is used by banks that are not sufficiently sophisticated.
- Standardized approach is similar to Basel I except for the calculation of risk weights.
 - Under Basel I, the headline risk weights depended on asset type and nationality of the obligor;
 - Under Basel II, the headline risk weights depended on obligor type and rating for some obligor types, and on asset type for others.
- Risk weights as a percent of principal

表 16: Risk Weights by Credit Rating

	AAA to AA -	A+ to A -	BBB+ to BBB -	BB+ to B -	Below B -	Unrated
Country	0	20	50	100	150	100
Banks	20	50	50	100	150	50
Corporations	20	50	100	100	150	100

- Example
 - Assets of Niu and Son's Bank includes \$100 million of bank's bond rated AA, \$200 million of government bond rated A+, and \$300 million of corporate bond rated BB-. Calculate riskweighted assets under Basel II standardized approach.
- Answer
 - $100 \times 20\% + 200 \times 20\% + 300 \times 100\% = 360$ million

16 Credit risk-mitigants

- Collateral: two approaches can adjust risk weights.
 - The first is termed the simple approach and is similar to an approach used in Basel I.
 - * For the part of the exposure covered by the collateral, the risk weight of the collateral is used.
 - * For any exposure not covered by the collateral, the risk weight of the counterparty is used.
 - * The minimum risk weight is 20% except for sovereign debt.

- Example
 - Red Star bank has \$100 million exposure to Giant company. The exposure is secured by \$90 million of T-bond rated AAA with a risk weight of 0%, and Giant company has a rating of A with a risk weight of 50%. Calculate risk-weighted assets applicable to the exposure using the simple approach.
- Answer
 - $90 \times 0 + 10 \times 50\% = 5$ million
- The second is termed the comprehensive approach.
 - Banks adjust the size of their exposure upward, and adjust the value of the collateral downward. The adjustments depend on the volatility of the exposure and the collateral.
 - A new exposure equal to the excess of the adjusted exposure over the adjusted value of the collateral.
- Example
 - Red Star bank has \$100 million exposure to Giant company. The exposure is secured by \$90 million of T-bond rated AAA with a risk weight of 0%, and Giant company has a rating of A with a risk weight of 50%. Exposure would increase 20% and value of collateral would decrease 5% in the future. Calculate risk-weighted assets under the comprehensive approach.
- Answer
 - New exposure: $100 \times 1.2 - 90 \times 95\% = 34.5$ million
 - RWA: $34.5 \times 50\% = 17.25$ million

17 Credit risk-foundation internal ratings based approach

- Capital requirement on VaR is calculated using a one-year time horizon and a 99.9% confidence level.
 - Expected loss is covered by financial products pricing.
 - Unexpected loss would be solved by VaR.
 - * Capital required is VaR minus expected loss.

18 Credit risk-foundation IRB approach

- Capital required:

$$\sum_i (EAD_i \times LGD_i \times DR_{99.9_i}) - EL$$

- ✓ EAD: The exposure at default (in dollars)
- ✓ LGD: The loss given default (expressed as a decimal)
- ✓ $DR_{99.9_i}$ or WCDR: The default rate at the 99.9th percentile for a large portfolio of assets of type i .
- ✓ EL: $\sum_i (EAD_i \times LGD_i \times PD)$

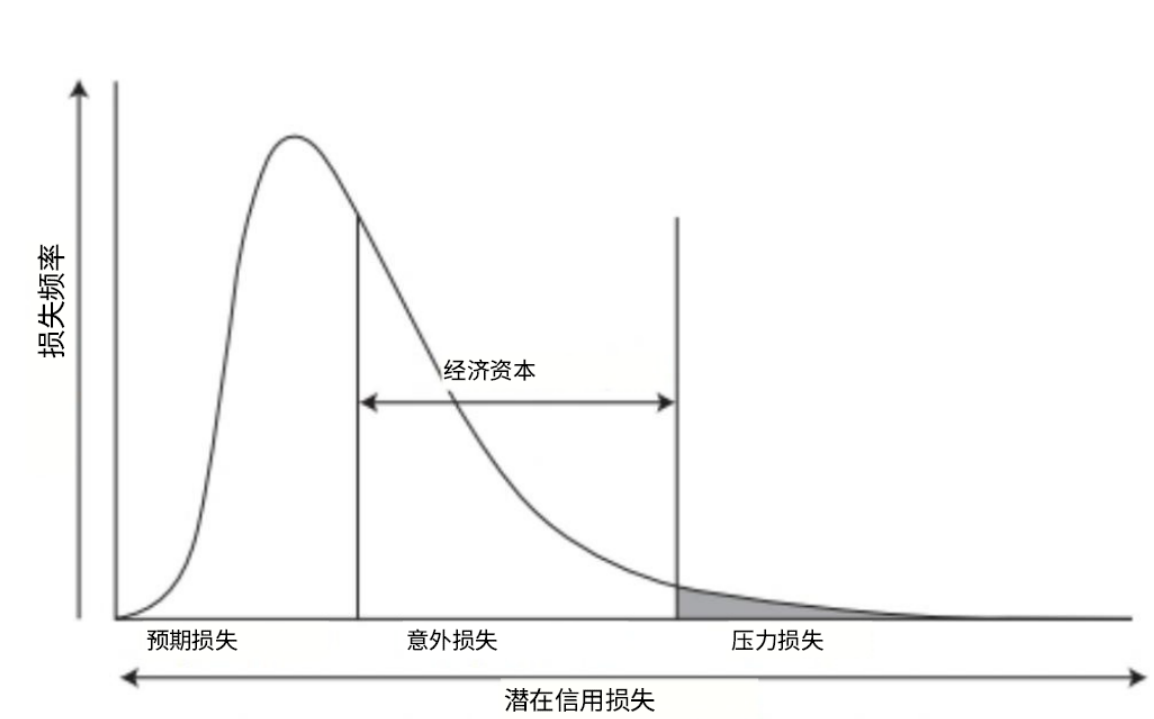


图 13: 潜在信用损失

- Loss distribution, expected and unexpected loss, and capital

- The final version of capital calculation:

$$RWA = 12.5 \times EAD \times LGD \times (DR_{99.9_i} - PD) \times MA$$

- MA is the maturity adjustment, equals $1 + \frac{(M-2.5) \times b}{1-1.5 \times b}$.
- M is the maturity of the exposure.

$$DR_{99.9_i} = N \left[N^{-1}(PD_i) - \sqrt{\rho} N^{-1}(0.001) / \sqrt{1 - \rho} \right] \quad (10)$$

- It is obvious that $DR_{99.9_i}$ increases as the copula correlation between each pair of obligors increases and as the probability of default increases.
- If the copula correlation is 0, $DR_{99.9_i}$ is equal to PD.

表 17: Values of PD for Different Correlation Coefficients

ρ	PD=0.001	PD=0.005	PD=0.01	PD=0.02
0	0.001	0.005	0.01	0.02
0.2	0.028	0.092	0.146	0.226
0.4	0.071	0.211	0.316	0.449
0.6	0.135	0.387	0.542	0.705

- PD has inverse relationship with correlation ρ .

$$\rho = 0.24 - 0.12 \times \left(\frac{1 - e^{-50PD}}{1 - e^{-50}} \right) \quad (11)$$

- As a company becomes less creditworthy, its PD increases and its probability of default becomes more idiosyncratic and less affected by overall market conditions.

- Under the Foundation IRB approach, banks supply PD while LGD, EAD, and M are supervisory values set by the Basel Committee.
 - The EAD is calculated similar to the credit equivalent amount (CEA) under Basel I. It includes the impact of netting.
 - MA is set to 2.5 in most cases.
- Under the advanced IRB approach, banks supply their own estimates of the PD, LGD, EAD, and M for corporate, sovereign, and bank exposures.
- For retail exposure, advanced IRB approach is used, but no maturity adjustment. Correlation are lower for retail than for wholesale exposure

19 Operational risk

- Three approaches to calculating capital for operational risk:
 - Basic indicator approach (simplest)
 - Standardized approach (middle)
 - Advanced measurement approach (most sophisticated)

20 Operational risk-standardized approach

- Basic indicator approach (BIA)

Capital calculation is a simple calculation of the average gross revenue for the past three years, multiplied by 15%.

- $K_{BIA} = \frac{\sum_{i=1}^n G_{I_i} \times \alpha}{n}$
 - GI: annual positive gross income
 - n: number of years in which gross income was positive
 - α : 15%

- Example

Niu and Son's Bank had annual gross revenue for the past three years as 200 million, -50 million, and 150 million respectively. Calculate the operational risk capital requirement.

- Answer

$$K_{BIA} = \frac{(200 + 150) \times 15\%}{2} = 26.25 \quad (12)$$

- Standardized approach (SA)

It captures operational risk factors, because different types of business activities carry different levels of operational risk.

- Beta factor(β) is a fixed percentage, which reflects the level of the gross income for each of the eight business lines.

- $$K_{TSA} = \frac{\sum_{years\ 1-3} \max[\sum(GI_{1-8} \times \beta_{1-8}), 0]}{n} \quad (13)$$

- GI_i : annual gross income in a given year

- beta factors

- 投行结算 (18%)
- 商行托管 (15%)
- 零经资产 (12%)

- Example

- Niu and Son's Bank had three lines of business for the past 3 years to do business (in \$million).
- Calculate the operational risk capital requirement.

表 18: Business Lines and Beta Factors

Business Lines	Beta Factors
Corporate finance (IB)	18%
Trading and sales (IB)	18%
Payment and settlement	18%
Commercial banking	15%
Agency (& custody) service	15%
Retail banking	12%
Retail brokerage	12%
Asset management	12%

表 19: Business Line Performance Over Three Years

Business Line	Year 1	Year 2	Year 3
Retail banking	10	10	20
Commercial banking	10	10	20
Settlement and payment service	-20	0	10

- Answer

- TSA capital charge per year:

表 20: Business Line Calculations Over Three Years

Business Line	Year 1	Year 2	Year 3
Retail banking	$10 \times 12\% = 1.2$	$10 \times 12\% = 1.2$	$20 \times 12\% = 2.4$
Commercial banking	$10 \times 15\% = 1.5$	$10 \times 15\% = 1.5$	$20 \times 15\% = 3.0$
Settlement and payment service	$-20 \times 18\% = -3.6$	$0 \times 18\% = 0$	$10 \times 18\% = 1.8$
Total	-0.9	2.7	7.2

–

$$K_{TSA} = \frac{2.7 + 7.2}{2} = 4.95 \text{ million} \quad (14)$$

- Advanced measurement approach (AMA) estimates distribution of losses in seven categories that incorporates estimates of both the incidence of operational loss events and their severity.
- Two broad approaches are most popular:
 - 1. A parametric and Monte Carlo approach: data parameterize the probability distribution for incidence/severity to combine, producing large simulated loss observations to the value in 99.9th.
 - 2. A scenario analysis approach: several detailed scenarios to measure operational losses in each scenario. Scenario analysis has the advantage of generating informative narratives and being forward-looking.
- Others
 - The capital requirement under AMA is based on one-year time horizon, 99.9% confidence level VaR.
 - Model must capture all expected and unexpected losses.
 - Four elements should be included, internal loss data, external loss data, scenario analysis, and business environment & control.
 - Bank must defend correlation assumptions in its AMA model.
 - Bank considers risk mitigating factors: may offset at most 20% of the operational risk capital charge with insurance compensation or claim instead of insurance premium.

21 Solvency II

- The long-standing regulatory framework for insurance companies in the European Union is known as Solvency I, is the process of being replaced by Solvency II.
- In the United States and the European Union, Solvency II establishes capital requirements for operational, investment, and underwriting risks of insurance companies.
- Solvency II is the "Basel II" for insurance company, and it has many similarities to Basel II.
 - Three Pillars: Pillar 1 quantitative requirement. Pillar 2 internal governance and official supervision. Pillar 3 disclosure and transparency.
 - Underwriting risk, credit and market risk, and operational risk are all considered. Underwriting risk is further detailed to risks arising from life insurance, P&C (property & casualty), health insurance.

22 Solvency II: SCR and MCR

- If its capital falls below SCR (solvency capital requirement) level, a restore capital plan should be delivered to supervisor. Supervisors force the insurance company to take particular measures to correct the situation.
- MCR (minimum capital requirement) is regarded as an absolute minimum level of capital. If capital drops below the MCR level, supervisors may prevent the insurance company from taking new business, or force the insurance company into resolution, transferring its policies to another company.

23 Solvency II: SCR calculating

- Calculating the SCR has two ways: standardized approach and internal models approach.
- Internal models approach is calculated with VaR concept with one-year and 99.5% confidence limit.
- Capital structure:
 - Tier 1 capital (equity, retained earnings and equivalents);
 - Tier 2 capital (liabilities subordinated to policyholders);
 - Tier 3 capital (subordinated to policyholders).

考点 2 Solvency, Liquidity and Other Regulation After the Global Financial Crisis

1 Basel II.5

- In the 2008 crisis, The Basel Committee recognized minimum capital under the market risk amendment were inadequate for the trading-book. It implemented updated rules by the end of 2011.
- It includes:
 - Stressed VaR
 - Incremental risk charge
 - Comprehensive risk capital

2 Market risk-stressed VaR

- Stressed VaR: Basel 11.5 requires banks to calculate two VaRs. One is the usual VaR (based on the previous one to four years of market movements). The other is stressed VaR .

- Stressed VaR is determined by basing calculations on how market variables moved during the one year (250-day) period from the most recent seven years that was most stressful for its current portfolio, rather than on how they moved during the past one to four years.
- The two VaR measures are combined to calculate capital charge.

$$\max(VaR_{t-1}, m_c \times VaR_{avg}) + \max(sVaR_{t-1}, m_s \times sVaR_{avg}) \quad (15)$$

- VaR_{t-i} and $sVaR_{t-i}$ are usual VaR and stressed VaR with 10-day time horizon, 99% confidence level on the previous day.
- VaR_{avg} and $sVaR_{avg}$ are the average of VaR and stressed VaR with 10-day time horizon, 99% confidence level over the previous 60 days.
- m_s and m_c are multiplicative factors that are determined by bank supervisors and are between 3 to 4.
- Example
Niu and Son's Bank calculated that the usual VaR for the previous day and average one covering the last 60 days are 18 million and 5.2 million, and the stressed VaR for the previous day and average one covering the last 60 days are 26.4 million and 11.3 million. Based on the backtesting result, Niu and Son's discovered the m_c is 3 and m_s is 3.5. Calculate the total market risk capital charge.
- Answer
 $\text{Max}(18, 5.2 \times 3) + \text{max}(26.4, 11.3 \times 3.5) = 18 + 39.55 = 57.55$

3 Market risk-IRC

- Incremental risk charge: the specific risk charge was intended to capture default and reaction to regulatory arbitrage opportunities between the banking and trading book. The Basel Committee named incremental default risk charge (IDRC)
 - the internal model of default risk calibrated to the same 99.9th percentile at a one-year horizon.
- Later, the committee had realized that most losses with credit risk due to changes in ratings, credit spreads, or liquidity, not defaults.
 - As a result, IDRC was widened to incremental risk charge (IRC).
- Comprehensive risk charge (CRC): An assumption in Basel II is that the correlation parameter is constant across obligors and over time. The assumption is wrong for instruments in the correlation book (securitizations/re-securitizations/ relating derivatives) .
- Such correlation-sensitive instruments put assets in SPV and create tranches differ in seniority. Correlations changes have large effects on the tranches.
 - The Basel Committee addressed the issue by replacing the IRC and specific risk charge with comprehensive risk (CR) charge for the correlation book. Comprehensive risk charge can use in either standardized approach or internal model approach.
- standardized approach

表 21: Standardized Capital Charge for Correlation-Dependent Instruments

Types of Instrument	AAA to AA-	A+ to A-	BBB+ to BBB-	BB+ to BB-	Below BB- or Unrated
Securitization	1.6%	4%	8%	28%	100%
Resecuritization	3.2%	8%	18%	52%	100%

- Internal model approach: If a bank is allowed to use an internal model, it must routinely perform rigorous stress tests.

4 Basel III

- After the global financial crisis, The Basal Committee submitted the proposals to remedy the deficiencies. The proposals were published in 2010 and 2011. It includes:
 - Capital requirements
 - Liquidity risk
 - Counterparty credit risk

5 Capital requirements

- Under Basel III, a bank's total capital consists of:
 - Tier 1 capital includes:
 - * Tier 1 equity capital
 - * Additional Tier 1 capital
 - Tier 1 capital excludes: Goodwill, other intangibles, deferred tax assets and any shortfall of reserves relative to IRB expected losses.
 - Tier 1 equity capital includes: common equity/retained earnings /a limited amount of minority interest and unrealized gains and losses
 - Additional Tier 1 capital: unsecured, unguaranteed, noncumulative perpetual preferred equity instruments subordinated to depositors and subordinated debt, and callable only after 5 years or more/debt with appropriate triggers that cause conversion to equity or write-downs/ approved minority interest not included in Core Tier 1
- Under Basel III, a bank's total capital consists of:
 - Tier 2 capital: protecting depositors and creditors after failure.
 - Subordinated debt: unsecured, unguaranteed debt instruments subordinated to depositors and subordinated debt, with 5 years or more original maturity, and callable only after 5 years or more.
 - General loan loss reserves: reserves not allocated to absorb losses on specific positions. Reserves included in capital are capped at 1.25% of standardized approach RWAs, or 0.6% of IRB RWAs
- Basel Committee refers common equity as "going-concern capital." When the bank is a going concern (i.e., has positive equity capital), common equity absorbs losses.
- Tier 2 capital is referred to as "gone concern capital." When the bank is no longer a going concern (i.e., has negative capital), losses have to be absorbed by Tier 2 capital.
- Tier 2 capital ranks below depositors in a liquidation. While Tier 2 capital remains positive, depositors should in theory be repaid.
- The minimum capital requirements are as follows:
 - 1. Tier 1 equity capital must be at least 4.5% of risk weighted assets at all times.
 - 2. Total Tier 1 capital (Tier 1 equity capital plus additional Tier 1 capital) must be at least 6% of risk-weighted assets at all times.
 - 3. Total capital (total Tier 1 plus Tier 2) must be at least 8% of risk-weighted assets at all times.

6 Capital requirements-buffers

- Capital conservation buffer (CCB): Basel III requires capital conservation buffer in normal times consisting of further amount of core Tier 1 equity capital equal to 2.5% of riskweighted assets

- It means the minimal value of Tier 1 equity capital ratio is 7%, that of Tier 1 capital is 8.5%, and that of total (Tier 1 plus Tier 2) is 10.5%.
- The rationale is that large buffer is specified to further reduce the chance of failure.
- Countercyclical capital buffer (CCB): countercyclical capital buffer that varies at the discretion of national supervisors and is between 0 and 2.5% of total risk-weighted assets and must be met with core Tier 1 equity capital assets.
- The rationales for CCB:
 - Provide for macroprudential restraint of overheating, higher capital requirements tend to restrict credit supply by banks.
 - The cost of capital: bank's cost of capital is smaller in good times than in bad times.

7 Capital requirements-G-SIBs buffer

- The Financial Stability Board(FSB) became the body in which many changes in international standards were approved.
- FSB publishes lists of G-SIB stands for global systemically important bank and G-SII for insurer, whereas the SIFIs means systemically important financial institutions.
- National regulators designate banks that have not been classified as G-SIBs as domestic systemically important banks (D-SIBs).
- The SI Fis are often presumed to be "too big to fail" and subjected to more regulation.
- Standard of SI Fis depends on the effect that its failure could have on the global financial system.
- G-SIBs are categorized according to whether the extra equity capital is 1%, 1.5%, 2%, 2.5%, or 3.5% of risk-weighted assets.
- Living wills: G-SIBs and D-SIBs are required to prepare detailed resolution plans when distressed, including how they would recapitalize and operate as a going concern.

Capital ratio requirements under Basel III			
Capital types	Tier 1 core or equity capital	Tier 1 capital	total capital
minimum capital requirements at all times	4.5%	6%	8%
Capital conservation buffer	2.5%		
New minimum value in normal time	7.0%	8.5%	10.5%
Countercyclical capital buffer	0~2.5% (非强制性要求, 由各国监管当局确定)		
Range of capital requirements	7%~9.5%	8.5%~11.0%	10.5%~13.0%
G-SIB(global systemically important bank)	1%~3.5%		
Example: JP Morgan Chase	9.5%~12%	11%~13.5%	13%~15.5%

图 14: CAPRAT

8 Capital requirements -leverage ratio

- Basel III specifies a minimum leverage ratio of 3% to supplement to the risk-based requirements.

- the leverage ratio is the ratio of a Tier 1 capital to leverage exposure.
 - Leverage exposure includes both on-balance-sheet assets and fractions of off-balance-sheet assets(e.g., derivatives or potential futures exposures).

9 Capital requirements- Cocos

- Contingent convertible bonds: increase equity automatically when distress occurs, as triggers in the indenture are touched.
- Common trigger is the ratio of Core Tier 1 Capital to RWA falls below a threshold, or regulator declares bank's nonviable.
- Cocos may be included in Additional Tier 1 or Tier 2.
- Economic benefits: As bonds, Cocos' cost is less than the cost of equity. Secondly, Cocos do not appear in the equity in normal time, a bank can report a higher ROE.

10 Liquidity risk

- Basel III has introduced requirements involving two liquidity ratios that are designed to ensure that banks can survive liquidity pressures.
- Liquidity coverage ratio (LCR)
- The bank has more liquid assets than it needs to meet a 30- day period of liquidity disruptions.
- Net stable funding ratio (NSFR)
- It focuses on liquidity management over a period of one year.

11 Liquidity risk: LCR

$$\text{Liquidity (LCR)} = \frac{\text{High-Quality Liquid Assets}}{\text{Net Cash Outflows in a 30-Day Period}} \quad (16)$$

- The 30-day period might include the following acute stress: a downgrade of the bank's debt by 3 notches (from AA to A), a partial loss of deposits, a complete loss of wholesale funding, increased haircuts on secured funding (collateral asset are not valued as highly), and drawdowns on lines of credit.
- Basel III regulations require the ratio to be greater than 100% so that the bank's liquid assets are sufficient to survive these pressures.

12 Liquidity risk: : NSFR

$$\text{Net Stable Funding Ratio (NSFR)} = \frac{\text{Amount of Stable Funding}}{\text{Required Amount of Stable Funding}} \quad (17)$$

- The numerator is calculated by multiplying each category of funding (liability and equity) by an available stable funding (ASF) factor, reflecting their stability.
- The denominator is calculated from the items requiring funding (asset). Each category of these is multiplied by a required stable funding (RSF) factor to reflect the permanence of the funding required.
- Basel III regulations require the ratio to be greater than 100%.

表 22: ASF Factor

ASF Factor	Category
100%	Tier 1 and Tier 2 capital, preferred stock and borrowing with a remaining maturity greater than one year
90%	“Stable” demand deposits and term deposits with remaining maturity less than one year provided by retail or small business customers
80%	“Less Stable” demand deposits and term deposits with remaining maturity less than one year provided by retail or small business customers
50%	Wholesale demand deposits and term deposits with remaining maturity less than one year provided by nonfinancial corporates, sovereigns, central banks, multilateral development banks, and public sector entities
0	All other liability and equity categories

表 23: RSF Factor

RSF Factor	Category
0%	Cash, short-term instruments, securities, loans to financial entities if they have a residual maturity of less than one year
5%	Marketable securities with a residual maturity greater than one year if they are claims on sovereign governments or similar bodies with a 0% risk weight
20%	Corporate bonds with a rating of AA- or higher and a residual maturity greater than one year, or claims on sovereign governments or similar bodies with a risk weight of 20%
50%	Gold, equity securities, bonds rated A+ to A-
65%	Residential mortgages
85%	Loans to retail and small business customers with a remaining maturity less than one year
100%	All other assets

- Example

- The balance sheet of Niu and Son’s Bank is following: (in %)

表 24: Assets and Liabilities

Asset	Value	Liability & Equity	Value
Cash	4	Wholesale demand deposits (<1 year)	34
Corporate bond (AAA)	16	Stable demand deposits (<1 year)	50
Gold	10	Preferred stock	5
Residential mortgage	60	Tier 2 capital	2.5
PPE	10	Tier 1 capital	8.5

- Answer
 - $ASF = 34 \times 50\% + 50 \times 90\% + 5 \times 100\% + 2.5 \times 100\% + 8.5 \times 100\% = 17 + 45 + 16 = 78$
 - $RSF = 4 \times 0 + 16 \times 20\% + 10 \times 50\% + 60 \times 65\% + 10 \times 100\% = 3.2 + 5 + 39 + 10 = 57.2$
 - $NSFR = \frac{78}{57.2} \approx 1.36$
 - It means Niu and Son's has met the requirement of NSFR.

13 Counterparty credit risk

- Credit value adjustment(CVA) for each derivatives counterparty is the difference in value between a risk-free portfolio of derivatives with that counterparty and the actual portfolio.
- Two approaches are available for calculating CVA risk: the standardized approach and basic approach.

14 Legislation and regulations after GFC

- Capacity to conduct macroprudential policy was added through institutional reforms in some nations where legal authority was previously lacking.
- Pre-crisis compensation at large banks that made pay effectively independent of risk-taking were widely blamed.
- Derivatives traded between financial institutions must be cleared by CCPs.
- In the United States, an Office of Credit Ratings was created at the SEC to provide oversight of rating agencies.
- In the United States, large banks were required to have board risk committees where at least one member has risk management experience.
- In the United States and the European Union, issuers of securitizations were required to retain at least 5 percent of each tranche, in an attempt to better-align the incentives of issuers and investors.

考点 3 High-Level Summary of Basel III Reforms

1 Motivations for revising the Basel III framework

- The Basel III framework is a central element of the Basel Committee's response to the global financial crisis.
- There are 3 motivations for revising Basel III framework
 - Addressing a number of shortcomings in the pre-crisis regulatory framework
 - Providing a foundation for a resilient banking system
 - Avoiding systemic vulnerability
- Initial phase of Basel III reforms focused on strengthening the following components of the regulatory framework:

- Improving the quality of bank regulatory capital by placing a greater focus on going-concern loss-absorbing capital in the form of Common Equity Tier 1 (CET1) capital
- Increasing the level of capital requirements to ensure banks are sufficiently resilient to withstand losses in times of stress.
- Enhancing risk capture by revising areas of the risk weighted capital framework that proved to be acutely miscalibrated.
- Adding macroprudential elements to the regulatory framework.
- specifying a minimum leverage ratio requirement to constrain excess leverage in the banking system and complement the risk-weighted capital requirements
- Introducing an international framework for mitigating excessive liquidity risk and maturity transformation, through the Liquidity Coverage Ratio and Net Stable Funding Ratio.
- The revision of Basel III seeks to restore credibility in the calculation of risk-weighted assets (RWAs) and improve the comparability of banks' capital ratios by 4 parts.
 - Enhancing the robustness and risk sensitivity of the standardized approaches for credit risk, credit valuation adjustment (CVA) risk and operational risk.
 - Constraining the use of the internal model approaches, by limits on certain inputs used to credit risk capital, removing the use of the internal model approaches for CVA risk and operational risk.
 - Introducing a leverage ratio buffer to further limit the leverage of global systemically important banks (G-SIBs).
 - Replacing the existing Basel II output floor with a more robust risk-sensitive floor based on the Committee's revised Basel III standardized approaches.

2 Dec 2017 revisions to Basel III framework

- December 2017 revisions to the Basel III framework in the following 5 areas:
 - Standardized approach to credit risk
 - Internal ratings-based (IRB) approaches for credit risk
 - CVA risk framework
 - Operational risk framework
 - Leverage ratio framework

3 Standardized approach for credit risk

- There are 8 parts on key revisions:
 - Unrated exposures
 - Risk weights for rated exposure
 - Exposure to corporates
 - Retail exposure
 - Residential real estate exposure
 - Commercial real estate exposure
 - Subordinated debt and equity exposure
 - Off-balance items
- The committee's revisions to the standardized approach for credit risk enhance the regulatory framework by:
 - Improving its granularity and risk sensitivity.

- Reducing mechanistic reliance on credit ratings, by requiring banks to conduct sufficient due diligence, and by developing a sufficiently granular non-ratings-based approach.
- Providing the foundation for a revised output floor to internally modelled capital and related disclosure to enhance comparability across banks and restore a level playing field.
- A more granular approach has been developed for unrated exposures to banks and corporates, and for rated exposures in jurisdictions where the use of credit ratings is permitted.
- For exposures to banks, some of the risk weights for rated exposures have been recalibrated. In addition, the riskweighted treatment for unrated exposures is more granular than the existing flat risk weight. A standalone treatment for covered bonds has also been introduced.
- For exposures to corporates, a more granular look-up table has been developed. A specific risk weight applies to exposures to small and medium-sized enterprises(SMEs). In addition, the revised standardized approach includes a standalone treatment for exposures to project finance, object finance and commodities finance.
- For retail exposures, a more granular treatment applies, which distinguishes between different types of retail exposures.
- For residential real estate exposures, more risk-sensitive approaches have been developed, whereby risk weights vary based on the LTV ratio of the mortgage(instead of the existing single risk weight) and in ways that better reflect differences in market structures.
- For commercial real estate exposures, approaches have been developed that are risk-sensitive than the flat risk weight which generally applies.
- For subordinated debt and equity exposures, a more granular risk weight treatment applies(relative to the current flat risk weight)
- For off-balance sheet items, the credit conversion factors(CCFs), which are used to determine the amount of an exposure to be risk-weighted, have been made more risksensitive.

4 IRB approaches for credit risk

- Internal ratings-based (IRB) approach has 3 shortcomings.
 - Excessive complexity of the IRB approaches
 - Lack of comparability in banks' internally modelled IRB capital requirements
 - Lack of robustness in modelling certain asset classes
- Committee has made 3 revisions to IRB approaches
 - Removing the option to use advanced IRB (A-IRB) approach for certain asset classes including exposures to large and mid-sized corporates, and exposures to banks and other financial institutions.
 - Adopting "input" floors to ensure a minimum level of conservatism in model parameters.
 - Providing greater specification of parameter estimation practices to reduce RWA variability.

表 25: Revised Scope of IRB Approaches for Asset Classes

Portfolio/Exposure	Basel II: available approaches	Basel III: available approaches
Large and mid-sized corporates (consolidated revenues > EUR 500 million)	A-IRB, F-IRB, SA	F-IRB, SA
Banks and other financial institutions	A-IRB, F-IRB, SA	F-IRB, SA
Equities	Various IRB approaches	SA
Specialised lending	A-IRB, F-IRB, slotting, SA	A-IRB, F-IRB, slotting, SA

5 CVA risk framework

- Credit value adjustment (CVA) could be used in initial phase of Basel III, as capital charge for potential mark-to-market losses of derivative instruments as a result of the deterioration in the creditworthiness of a counterparty.
- Committee revises CVA framework in 3 parts:
 - Enhance its risk sensitivity
 - Strengthen its robustness
 - Improve its consistency

6 Operational risk framework

- The Committee has streamlined the operational risk framework.
- All existing measurement approaches, including basic indicator approach, standardized approach, alternative standardized approach and advanced measurement approaches (AMA) are replaced with a single risk sensitive standardized approach (or standardized measurement approach, SMA) to be used by all banks.

7 Leverage ratio framework

- Leverage ratio complements the risk-weighted capital requirements. To maintain the incentives of capital constraints, the finalized Basel III reforms introduce a leverage ratio buffer for G-SIBs.
- The leverage ratio G-SIB buffer must be Tier 1 capital and set at 50% of G-SIB's risk weighted higher-loss absorbency requirements.

表 26: Capital conservation ratios for a G-SIB subject to a 1% risk-weighted buffer and 0.5% leverage ratio buffer

CET1 risk-weighted ratio	Tier 1 leverage ratio	Minimum capital conservation ratios (expressed as a percentage of earnings)
4.5–5.375%	3–3.125%	100%
> 5.375–6.25%	> 3.125–3.25%	80%
> 6.25–7.125%	> 3.25–3.375%	60%
> 7.125–8%	> 3.375–3.50%	40%
> 8.0%	> 3.50%	0%

Excerpts from High-level summary of Basel III reforms, BCBS, www.bis.org

8 Review output floor

- The Basel III reforms replace the existing Basel II floor with a floor based on the revised Basel III standardised approaches.
- The revised floor places limits on the regulatory capital in internal models relative to the standardized approaches to maintain a level playing field and comparability between banks.
- Under the revised output floor, bank's risk-weighted assets must be calculated as the higher of
 - 1) total risk-weighted assets using approaches in accordance with the Basel capital framework (both standardized and internal model-based approaches);
 - 2) 72.5% of total risk-weighted assets using only standardized approach.

表 27: RWA Calculation

风险类型	使用整体底线前的 RWA	标准法计算的 RWA	标准法计算的 RWA 的 72.5%
信用风险	62	124	89.9
其中: A 类资产	45	80	58
B 类资产	5	32	23.2
C 类资产	12	12	8.7
市场风险	2	4	2.9
操作风险 (非模型法)	12	12	8.7
风险加权资产合计	76	140	101.5

考点 4 Basel III: Finalising Post-Crisis Reforms

1 SMA: operation risk capital charge

- Standardized approach, or standardized measurement approach (SMA) measures minimum operational risk capital requirements, and could replace all existing approaches in the Basel II framework.
- SMA is based on the following 3 components:
 - Business indicator (BI), financial statement-based proxy
 - Business indicator component (BIC), equaling BI multiplied by a set of regulatory determined marginal coefficients (α_i)
 - Internal loss multiplier (ILM), a scaling factor based on a bank's average historical losses and the BIC

2 SMA: business indicator (BI)

$$BI = ILDC_{avg} + SC_{avg} + FC_{avg} \quad (18)$$

- BI is derived from average data from three years.
- $ILDC = \min[\text{Abs}(\text{interest income} - \text{interest expense}); 2.25\% \times \text{income earning assets}] + \text{dividend income}$
- $SC = \max[\text{other operating income}; \text{other operating expense}] + \max[\text{fee income}; \text{fee expense}]$
- $FC = \text{Abs}(\text{net P\&L trading book}) + \text{Abs}(\text{net P\&L banking book})$

3 SMA: business indicator component (BIC)

- Example:

The marginal coefficients increase with the size of the BI. Given Claire bank has BI 3Sbn Euros, Calculate the bank's BIC.

表 28: BI Buckets and Marginal Coefficients

BI Bucket	BI Range	Marginal BI Coefficients (α_i)
1	$\leq 1\text{bn euros}$	0.12
2	$1\text{bn euros} < BI \leq 30\text{bn euros}$	0.15
3	$> 30\text{bn euros}$	0.18

- Answer:
- $1 \times 12\% + (30-1) \times 15\% + (35-30) \times 18\% = 5.37\text{bn Euros}$

4 SMA: internal loss multiplier (ILM)

- Through internal loss multiplier (ILM), a bank's internal operational risk loss experience affects the operational risk capital.
 - ILM could be calculated by the inputs of both
 - * loss component (LC): 15 times average annual operational risk losses incurred over previous 10 years
 - * business indicator component (BIC)
 - * $ILM = \ln \left[e^1 - 1 + \left(\frac{LC}{BIC} \right)^{0.8} \right]$

5 SMA: operational risk capital requirement

- Operational risk capital requirement is determined by the product of the BIC and the ILM.
 - Minimal operational risk capital (ORC) = BIC * ILM
- For banks in bucket 1 (with BI < 1bn Euros), internal loss data does not affect the capital calculation. That is, ILM=1, so the operational risk capital is equal to the BIC = 12%* BI
- At national discretion, supervisors may set the value of ILM equal to 1 for all banks in their jurisdiction.
- Banks which do not meet the loss data standards are required to hold capital that is at a minimum equal to 100% of the BIC. Supervisors may require the bank to apply an ILM which is greater than 1.

6 SMA vs. earlier approach

- Standardized approach (SA), alternative standardized approach (ASA) and advanced measurement approach (AMA) are currently required to collect operational losses and report these losses to supervisors.
 - SMA is a single, non-model-based method.
 - AMA is able to use a range of alternative models

7 Loss data identification, collection and treatment

- There are 8 items on gross criteria on loss data identification, collection and treatment.
 - 1. Internally generated loss data calculations used for regulatory capital purposes must be based on a 10-year observation period.
 - 2. Internal loss data are most relevant when clearly linked to a bank's current business activities, technological processes and risk management procedures.
 - 3. For risk management purposes, and to assist in supervisory validation and/or review, a supervisor may request a bank to map its historical internal loss data into the Basel II Framework and to provide this data to supervisors.
 - 4. A bank's internal loss data must be comprehensive and capture all material activities and exposures from all appropriate subsystems and geographic locations.
 - 5. Aside from information on gross loss amounts, the bank must collect information about the reference dates of operational risk events.
 - 6. Operational loss events related to credit risk and that are accounted for in credit risk RWAs should not be included in the loss data set.
 - 7. Operational risk losses related to market risk are treated as operational risk for the purposes of calculating minimum regulatory capital under this framework and will therefore be subject to the standardized approach for operational risk.

- 8. Banks must have processes to independently review the comprehensiveness and accuracy of loss data.
- Specific criteria on loss data identification, collection and treatment.
 - Building of the standardized approach loss data set
 - * Building an acceptable loss data set from the available internal data requires that the bank develop policies and procedures to address several features, including gross loss definition, reference date and grouped losses.
- Gross loss, net loss, and recovery definitions Included:
 - 1.Direct charges, including impairments and settlements, to the bank’s P&L accounts and write-downs due to the operational risk event;
 - 2.Costs incurred as a consequence of the event including external expenses with a direct link to the operational risk event and costs of repair or replacement, incurred to restore the position that was prevailing before the operational risk event;
 - 3.Provisions or reserves accounted for in the P&L against the potential operational loss impact;
 - 4.Losses stemming from operational risk events with a definitive financial impact, which are temporarily booked in transitory and/or suspense accounts and are not yet reflected in the P&L. Material pending losses should be included in the loss data set within a time period commensurate with the size and age of the pending item; and
 - 5.Negative economic impacts booked in a financial accounting period, due to operational risk events impacting the cash flows or financial statements of previous financial accounting periods. Material ”timing losses” should be included in the loss data set when they are due to operational risk events that span more than one financial accounting period and give rise to legal risk.
- Gross loss, net loss, and recovery definitions. Excluded:
 - 1.Costs of general maintenance contracts on PP&E;
 - 2.Internal or external expenditures to enhance the business after the operational risk losses: upgrades, improvements; and
 - 3.Insurance premium.

考点 5 Supplementary (FRTB)

1 FRTB

- In May 2012, Basel Committee on Banking Supervision issued a consultative document: Fundamental Review of the Trading Book (FRTB), proposing major revisions to the way capital for market risk is calculated for the trading book. The final version was under the title: Minimum capital requirements for market risk
 - Background
 - Liquidity horizons
 - Market risk capital charge

2 Background

- Trading Book VS. Banking Book
 - Trading Book consist of instruments that the bank intends to trade
 - Banking Book consists of instruments that are expected to be held to maturity. Instruments in the banking book are subject to credit risk capital whereas those in the trading book are subject to market risk capital. This has in the past given rise to regulatory arbitrage.

- Market risk capital in Basel I is based on VaR for a 10-day horizon with a 99% confidence level, the VaR were based on the behavior of market variables during an immediately preceding period of time(typically, one to four years).
- FRTB is proposing a change to the measure for determining market risk capital.
 - Instead of VaR with a 99% confidence level, expected shortfall (ES) with a 97.5% confidence level is proposed, actually stressed ES with a 97.5% confidence level.
 - If losses have a normal distribution that has a mean μ and standard deviation σ , 99% VaR is $\mu + 2.326\sigma$, 97.5% ES is $\mu + 2.338\sigma$, the two measures are almost exactly equivalent.
 - For non-normal distributions, they are not equivalent. When the loss distribution has a heavier tail than a normal distribution, the 97.5% ES can be considerably greater than the 99% VaR.

3 Liquidity horizons

- Under FRTB, the 10-day time horizon used in Basel I and Basel II.5 is changed to reflect the liquidity of the market variable being considered.
- FRTB considers changes to market variables that would take place(in stressed market conditions)over periods of time reflecting their liquidity.
- Liquidity horizon represents the time required to sell the position or to hedge all material risks in a stressed market.
 - It reflects the differing liquidities of market variables.
 - Five different liquidity horizons are used: 10 days, 20 days, 40 days, 60 days, and 120 days.
 - Both internal models-based and standardized approaches are accepted.

4 Market risk capital charge

- The Basel III has specified a standardized approach and an internal models approach.
 - Even the bank has been approved by their supervisor to use the internal models approach, banks must also implement the standardized approach.
- Standardized Approach: The capital requirement is the sum of three components:
 - A risk charge calculated using a risk sensitivity approach
 - A default risk charge
 - A due residual risk add-on
- A risk charge calculated using a risk sensitivity approach
 - Seven risk classes(corresponding to trading desks)
 - * General interest rate risk, foreign exchange risk, commodity risk, equity risk, three categories of credit spread risk.
 - * Within each risk class, a delta risk charge, vega risk charge, curvature risk charge are calculated.
- Delta risk charge measures the sensitivity of the portfolio to percentage changes of the prices of underlying assets.
- Vega risk charge measures the sensitivity of the value of the portfolio to small changes in volatility.
- Curvature risk charge is a capital charge for a bank's gamma risk exposure under the standardized approach.
- Default risk charge
 - Credit spread risks: handled using the delta/vega/curvature approach in the standardized approach.
 - Default risks: sometimes referred to as jump-to-default(JTD) risk, are handled by a separate default risk charge. Calculated by multiplying each exposure by LGD and default risk weight.
 - * Both the LGD and risk weight are specified by the Basel committee.

- Residual risk add-on: risks that cannot be handled by the delta/vega/curvature approach.
 - It includes exotic options when they cannot be considered as linear combinations of plain vanilla options.
 - The add-on is calculated by multiplying the notional amount of the transaction by a risk weight that is specified by the Basel committee.
- Internal Models Approach:
 - It requires banks to typically use the historical simulation approach to estimate stressed ES with a 97.5% confidence, covering the different liquidity horizon of 10 days, 20 days, 40 days, 60 days, and 120 days.
- Credit risk:
 - Credit spread risk: the risk that credit spread changes causing the mark-to-market value of the instrument to change. It handled in a similar way to other market risks.
 - Jump-to-default risk: the risk that there will be a default by the company. It is handled in the same way as default risks in the banking book based on VaR calculation with oneyear time horizon and 99.9% confidence level.
- Securitization:
 - The comprehensive risk measure (CRM) charge was introduced in Basel 11.5 to cover the risks in products created by securitizations such as ABS/CDO. The CRM rules allow a bank (with regulatory approval) to use its own model.
 - The Basel Committee realized this is unsatisfactory because there is too much variation by different banks for same portfolio.
 - It decided that under FRTB, the standardized approach must be used for securitizations.

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